

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)



ANAIS
(Autobox and Autophoria
Inventory System)

Version 0.4

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AN AIS	2/24/26	• Completion of SRS 1 first draft	V 0.1.0
AN AIS	03/02/26	• Fixing of minor errors • Adding of signatures • Additional Diagrams	V 0.2
AN AIS	03/15/26 to 03/25/26	• Revision and enhancement of Section 1.1 (Purpose) to conform with IEEE SRS documentation standards • Update of Section 1.2 (Scope) to incorporate the Supplier Management module and Automated Supplier Ordering feature • Addition of new terms to Section 1.3 (Definitions): Purchase Order, Supplier, Stock-Out • Addition of new acronym to Section 1.3: PO (Purchase Order) • Update of Section 3.1 (Product Perspective) context diagram – added Supplier as an external entity with Auto-generated PO and Product Delivery interactions; added Draft PO Notification flow to Owner • Update of Section 3.2 (Product Functions) table and diagram – added Manage Supplier, Record Purchase Order, Record Delivery,	V 0.3

		and Automated PO Generation functions • Update of Section 3.6 (Apportioning of Requirements) – Supplier Management and Automated Supplier Ordering moved to Initial Release; updated deferred requirements list • Completion of Section 4.1 Functional Requirements: 16 structured use case specifications (UC01-UC16), including Supplier Management (UC13-UC15) and Automated Purchase Order Generation (UC16) • Update of UC10 (Receive Low-Stock Alert) Extends field updated to reference UC16 • Completion of Section 4.2 System Use Case Diagram updated to include UC16 with «extends» from UC10 and System actor	
AN AIS	04/06/2025 to		V 0.4

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines and documents the complete functional and non-functional requirements of the Autobox and Autophoria Inventory System (AN AIS), Version 0.2.1. AN AIS is a centralized, web-based inventory management system developed for Autobox (La Trinidad, Benguet) and Autophoria (Ferguson, Baguio City), two automotive accessory branches under the sole management of Rodny Castillo.

This document serves as the authoritative reference for the design, development, testing, and deployment of AN AIS throughout the project lifecycle. It is intended to provide a clear, complete, and unambiguous specification of system behavior and requirements for all project stakeholders, including the business owner, branch officers-in-charge (OICs), branch employees, the student development team, and the academic project reviewer.

This SRS establishes the scope of the system, defines the roles and access levels of all users, and specifies the detailed functional requirements governing inventory tracking, stock management, supplier coordination, purchase order processing, and consolidated reporting across both branch locations. It shall be used as the primary basis for system development and shall be referenced throughout all phases of the project to ensure that the delivered system aligns with the identified business objectives and stakeholder expectations.

1.2 Scope

The Autobox and Autophoria Inventory System (AN AIS) is a centralized, web-based platform designed to manage and monitor inventory operations across both branch locations. The system provides real-time, branch-based inventory tracking, stock movement monitoring, supplier and purchase order management, sales recording, and consolidated reporting. AN AIS enables the owner to oversee stock levels and operational performance across both branches from a unified interface, while authorized

branch staff manage daily inventory and sales transactions within their defined access permissions.

The system is focused specifically on inventory and stock control processes, including supplier coordination and purchase order tracking. ANAIS will not include advanced accounting modules, payroll systems, customer relationship management (CRM), or e-commerce integration in its current version.

The Autobox and Autophoria Inventory System (ANAIS) is limited to:

- Landing Page
- Log-In / Log-Out System
- Role-based Account Creation and Management (limited to the Owner)
- Product Record Management – adding, editing, and deleting product records (limited to the Owner and OIC)
- Stock-In Recording – recording of incoming product deliveries (Owner and OIC)
- Stock-Out Recording – recording of product sales transactions (Owner and OIC)
- Supplier Record Management – registering and maintaining supplier information (limited to the Owner)
- Purchase Order Management – creating and tracking purchase orders from registered suppliers (limited to the Owner)
- Supplier Delivery Recording – recording of actual product receipts against purchase orders (Owner and OIC)
- Viewing of product availability, inventory levels, and stock status (all user roles)
- Inventory Report Generation – branch-wise and consolidated reports (Owner and OIC)
- Low-Stock Alert Notifications (Owner and OIC)

The system is intended for use by the following user roles:

- Owner of the business (Administrator)
- Co-Owner of the business
- Officer-in-Charge (OIC) for each branch
- Employees / Staff of both Autobox and Autophoria
- Automated purchase order creation triggered by low-stock threshold alerts (limited to the Owner)

User access is divided as follows:

- Owner
 - Full system access – all functions including product management, supplier management, purchase order creation, stock-in/out recording, report generation, and account management
 - Oversees inventory operations across both branches
- Co-Owner
 - Provides input on stock control decisions
 - Reviews system functionality and monitors inventory reports
- Officer-in-Charge (OIC)
 - Records stock-in and stock-out transactions
 - Records supplier deliveries against purchase orders
 - Monitors stock levels and generates reports
 - View-only access for supplier and product records
- Employee
 - View-only access – may view product availability, inventory levels, and search for products

1.3 Definitions, Acronyms, and Abbreviations.

Definitions

- **Administrator / Owner** – The business owner (Rodny Castillo) with full access and control over all system functions, including product, supplier, purchase order, report, and user account management.
- **Co-Owner** – An additional business stakeholder who may provide input on stock control decisions, review inventory reports, and oversee operational performance. Has limited administrative rights compared to the Owner.
- **Employee** – Staff such as sales personnel, lubemen, mechanic helpers, and installers; has view-only access to the system unless assigned specific tasks by the Owner.
- **Officer-in-Charge (OIC) / Transaction Manager** – Branch-level staff responsible for recording stock-in, stock-out, and supplier delivery transactions; can view reports and monitor inventory.

- **Inventory** – A complete record of all products, spare parts, and supplies available at each branch, including quantities, categories, and current stock status.
- **Stock-In** – The process of adding received products to inventory, either through general stock entry or supplier delivery recording against a purchase order.
- **Stock-Out** – The process of deducting product quantities from inventory due to a sales transaction.
- **Purchase Order (PO)** – A formal document created by the Owner to request specific products from a registered supplier. Includes line items, quantities, unit costs, expected delivery date, and status (Pending, Partially Delivered, Fully Delivered, Cancelled).
- **Supplier** – An external business or individual providing products. Must be registered in ANAIS before being referenced in purchase orders; suppliers have no login access.
- **Invoice** – A printed or digital document issued to a customer as proof of a completed sales transaction.
- **Walk-in Customer** – A customer who visits the branch without prior appointment or reservation.
- **Low-Stock Alert** – An automated system notification triggered when a product's quantity reaches or falls below the predefined reorder threshold.
- **Dashboard** – A system interface displaying real-time inventory status, low-stock alerts, and branch performance data based on user role.
- **Role-Based Access Control (RBAC)** – A system feature restricting user access to functions and data based on assigned roles (Owner, OIC, Employee).
- **Centralized System** – A system where all data is stored in a single database accessible by both branches.
- **Real-Time Processing** – Immediate updating of inventory data upon any transaction.
- **Data Integrity** – Ensuring accuracy and consistency of inventory and transaction data over time.
- **Authentication** – The process of verifying a user's identity through login credentials.
- **Authorization** – Granting access rights and permissions based on user role.
- **Session** – A period during which a user is logged into the system.
- **CRUD Operations** – Core functions of the system: Create, Read, Update, Delete records.

- **Report Generation** – Creating branch-wise or consolidated reports for monitoring stock levels, sales, and operational decisions.
- **System Backup** – Routine saving of database records to prevent data loss.

Acronyms

- **AN AIS** – Autobox and Autophoria Inventory System
- **API** – Application Programming Interface
- **BIR** – Bureau of Internal Revenue
- **CRUD** – Create, Read, Update, Delete
- **OIC** – Officer-in-Charge
- **PO** – Purchase Order
- **POS** – Point of Sale
- **RBAC** – Role-Based Access Control
- **SKU** – Stock Keeping Unit

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1.5 Overview

This Software Requirements Specification (SRS) defines the detailed requirements for the Autobox and Autophoria Inventory System (AN AIS), a centralized, web-based inventory management solution designed to support the day-to-day operations of the two automotive accessory branches owned by Rodny Castillo. AN AIS aims to streamline inventory control, stock movement tracking, supplier and purchase order management, sales recording, and consolidated reporting across both branches, providing real-time visibility and operational oversight for the owner while enabling branch staff to efficiently manage daily inventory tasks.

The remainder of this SRS is structured as follows:

- **Section 2: Business Description** - Provides the context for the system by describing the background of Autobox and Autophoria, including their organizational structure, business objectives, and stakeholder profiles. This section explains the business challenges AN AIS is intended to address and the justification for its development.
- **Section 3: Overall Description** - Details the system perspective, operational environment, product functions, user characteristics, constraints, assumptions, and dependencies that shape the design and implementation of AN AIS. It also describes system interfaces, hardware and software requirements, and operational considerations.
- **Section 4: Specific Requirements** - Specifies the functional requirements of AN AIS in detail, including user interactions, workflows, and use case diagrams. This section provides developers with the precise behavior and expectations of the system.
- **Section 5: Supporting Requirements** - Covers interface designs, logical data structures, and additional technical specifications that support system implementation and ensure the requirements are met effectively.

2 Business Description

2.1 Business Description

Autobox, located in La Trinidad, and Autophoria, situated in Baguio City, are privately owned automotive accessory shops managed by Rodny Castillo. Both establishments operate within the automotive retail industry, offering a wide range of vehicle accessories including car audio systems, lighting components, interior and exterior enhancements, and performance parts. The businesses cater to a growing market of vehicle owners seeking customization, improved functionality, and enhanced driving experiences.

Since their establishment, Autobox and Autophoria have built a steady customer base through quality products and personalized services. As the businesses expanded into operating multiple branches, the complexity of managing inventory across locations also increased. Currently, both branches rely on manual and traditional inventory practices, such as handwritten records and basic monitoring methods. While these approaches have been functional during the early stages of the business, they have become increasingly inefficient due to the high volume and fast turnover of products.

The lack of a centralized and automated system makes it difficult to track real-time stock levels, identify fast-moving and slow-moving items, and maintain accurate inventory records across both branches. This often results in issues such as stock discrepancies, delayed restocking, overstocking of certain items, and stockouts of high-demand products, ultimately affecting sales performance and customer satisfaction.

To address these challenges, the proposed Autobox and Autophoria Inventory System (AN AIS) is designed to modernize and streamline inventory management processes. The system will provide real-time monitoring of stock levels, automated tracking of inventory movement, and centralized data access for

both branches. By integrating inventory information into a unified digital platform, ANAIS enhances operational efficiency, improves coordination between locations, and supports informed decision-making for the business owner.

Given the nature of the business, which involves handling a diverse range of automotive products with high turnover rates, implementing an automated inventory system is essential. ANAIS directly responds to the operational needs of the organization by ensuring accurate inventory control, reducing human error, and enabling timely restocking. This ultimately helps maintain product availability, optimize stock levels, and improve overall business performance.

2.2 Business Objectives

The Autobox and Autophoria Inventory System (ANAIS) aims to deliver a practical, centralized, and efficient inventory management solution that supports the operational needs of both branches while remaining feasible within the project scope and timeline.

1. To develop and deploy a centralized inventory system for Autobox in La Trinidad and Autophoria in Baguio City within the academic project timeline, enabling unified access to inventory data across both locations.
2. To improve stock accuracy through real-time tracking of incoming and outgoing products, minimizing errors caused by manual recording and enhancing data reliability.
3. To reduce inventory discrepancies between branches by implementing synchronized and centralized data access, ensuring consistency of stock records.
4. To generate branch-wise and consolidated inventory reports on demand, enabling the business owner and staff to make informed decisions regarding restocking, product demand, and inventory distribution.
5. To ensure system feasibility and usability by utilizing existing hardware and software resources and conducting staff training, allowing employees to effectively use the system

within a short period after deployment.

6. To maintain a focused and achievable system scope by limiting ANAIS to inventory and stock control functions, including product management, stock-in and stock-out recording, supplier management, and purchase order tracking, while excluding modules such as accounting, payroll, CRM, and e-commerce integration, ensuring successful completion within the given timeframe.

2.3 Stakeholder Profile

Owner - Autobox & Autophoria

Representative	Rodny Castillo
Description	Primary business owner overseeing both branches
Type	Business
Responsibilities	Approve system requirements, provide business rules, make final decisions regarding inventory policies
Involvement	Decision-maker, requirements reviewer, provides feedback
Deliverables	Access to branch data
Comments / Issues	Needs the system to be simple and easy to use; may request changes if reports are unclear or interface is complex

Co-Owner

Representative	Ellen Castillo
Description	Co-owner assisting with business oversight
Type	Business
Responsibilities	Review system functionality, monitor inventory reports, provide input on stock control
Involvement	Requirements reviewer, validation
Deliverables	Feedback on reports and system usability
Comments / Issues	Possible differences in priorities with primary owner; may require clarification on roles

OIC - Autophoria

Representative	Crisanto Oreiro
Description	Branch Manager for Autophoria

Type	Expert user
Responsibilities	Daily inventory updates, monitor stock, update stock, report issues to owner
Involvement	Daily inventory updates, monitor stock, update stock, report issues to owner
Deliverables	Daily inventory records, low-stock alerts
Comments / Issues	May need extra guidance or training to use the system efficiently; staff turnover could affect usage

OIC - Autobox

Representative	Marcko Carbonel
Description	Branch manager for Autobox
Type	Expert user
Responsibilities	Daily inventory updates, monitor stock, update stock, report issues to owner
Involvement	End-user, requirements contributor, tester
Deliverables	Daily inventory records, low-stock alerts
Comments / Issues	May have limited computer skills; could request simpler interface or step-by-step instruction

OIC - Autobox

Representative	Jonathan Soriano
Description	Branch manager for Autobox
Type	Expert user
Responsibilities	Assist with stock management, validate inventory data, support staff
Involvement	End-user, requirements contributor
Deliverables	Inventory transaction confirmations
Comments / Issues	May overlap responsibilities with other OIC; needs clear role definition

Project Instructor

Representative	Martinez, Erna Kristi
Description	Academic supervisor for the student project
Type	Expert
Responsibilities	Guide project methodology, review deliverables, ensure learning outcomes
Involvement	Reviewer, mentor, validator
Deliverables	Evaluation reports, feedback on SRS

Comments / Issues	May require revisions to meet academic standards or grading criteria
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Student Team Member

Representative	Albino, Erin Camille
Description	Student developer
Type	Casual user / developer
Responsibilities	Design, develop, and implement system modules
Involvement	Developer, tester, documenter
Deliverables	Code, documentation, test results
Comments / Issues	Limited experience may affect development speed, requires coordination with team

Student Team Member

Representative	Arjona, Clifford Xhyrus
Description	Student developer
Type	Casual user / developer
Responsibilities	Design, develop, and implement system modules
Involvement	Developer, tester, documenter
Deliverables	Code, documentation, test results
Comments / Issues	May need peer support for complex coding tasks, time management is critical

Student Team Member

Representative	Cauan, Nel Angelo
Description	Student developer
Type	Casual user / developer
Responsibilities	Design, develop, and implement system modules
Involvement	Developer, tester, documenter
Deliverables	Code, documentation, test results
Comments / Issues	Must manage version control and integration carefully, risk of conflicting updates

Student Team Member

Representative	Lopez, Hans Jayden
Description	Student developer
Type	Casual user / developer

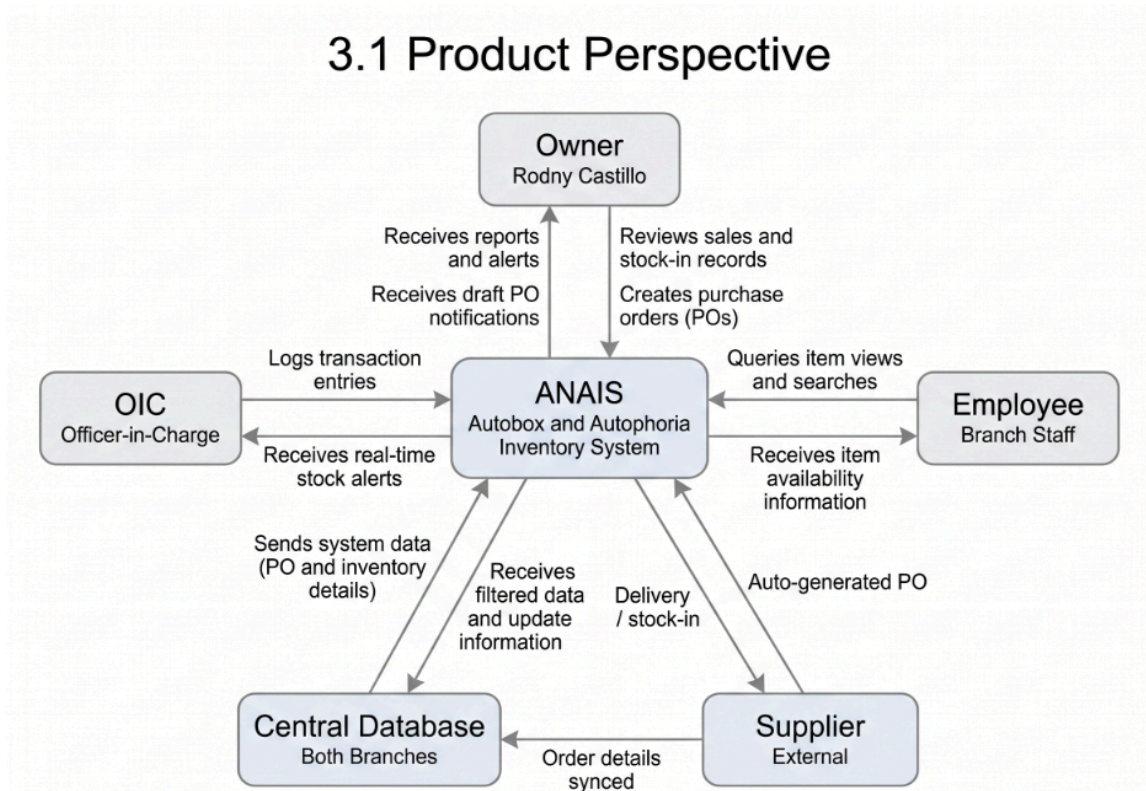
Responsibilities	Design, develop, and implement system modules
Involvement	Developer, tester, documenter
Deliverables	Code, documentation, test results
Comments / Issues	Needs to ensure system modules are integrated correctly with other team members' work

Student Team Member

Representative	Zapanta, Zeus Levi
Description	Student developer
Type	Casual user / developer
Responsibilities	Design, develop, and implement system modules
Involvement	Developer, tester, documenter
Deliverables	Code, documentation, test results
Comments / Issues	Must ensure interface is user-friendly for branch staff, limited prior experience may require guidance

3 The Overall Description

3.1 Product Perspective



The Autobox and Autophoria Inventory System (ANAIS) is a centralized, web-based inventory management system developed for Autobox La Trinidad and Autophoria Baguio City. It is a self-contained system designed specifically to manage and synchronize inventory operations between the two branches.

ANAIS replaces manual and spreadsheet-based inventory tracking with a unified database system accessible to authorized users via a web browser. ANAIS processes inventory inputs, updates stock in real time, and generates reports for both branches.

The system interacts with the following external users and entities:

- **Owner** - Views consolidated and branch-wise inventory reports; manages products, suppliers, purchase orders, and user accounts.

- **Co-Owner** - Provides input on stock control decisions, reviews inventory reports, and monitors operational performance. Has limited administrative rights compared to the Owner.
- **Branch Staff / Officer-in-Charge (OIC)** - Records sales transactions, updates stock levels, and records supplier deliveries.
- **Employee** - Views product availability and performs search and lookup requests within the system.
- **Supplier (external entity)** - Receives auto-generated purchase orders from the system when stock levels are low, and provides delivery and stock-in data upon order fulfillment.

3.1.a System Interfaces

The Autobox and Autophoria Inventory System (AN AIS) operates independently and does not interface with external enterprise systems such as payroll, accounting, CRM, or e-commerce platforms. The system uses a centralized internal database shared by both branches. Communication between the user interface and the backend is handled via internal APIs, which process requests for inventory management tasks such as viewing stock levels, recording stock-in/out transactions, managing product and supplier records, and generating reports. This ensures organized data flow, maintainability, and real-time updates across both branches.

3.1.b User Interfaces

AN AIS provides a Graphical User Interface (GUI) accessible through a web browser. The system is designed for non-technical users with basic computer knowledge.

Forms for:

- Stock-in (deliveries)
- Stock-out (sales)
- Product management
- Report generation page

Interface Optimization:

- Simple and clean layout
- Clear labels and navigation menus

- Minimal technical terminology
- Consistent button placement
- Basic accessibility support (readable fonts, clear contrast)

3.1.c Hardware Interfaces

AN AIS interacts only with:

- Desktop computers
- Keyboard and mouse

3.1.d Software Interfaces

The system uses the following required software components:

- **Web Browser** - Purpose: User access interface.
Interface: HTTP/HTTPS request-response format.

3.1.e Communications Interfaces

AN AIS operates over:

- WAN (Wide Area Network) - e.g., HTTPS, FTP, SMTP
- LAN (Local Area Network)

3.1.f Memory Constraints

There are no strict memory constraints. The system is designed to run on standard office computers with typical business specifications. No high-memory processing is required.

3.1.g Operations

Normal Operations:

- **Daily Stock Entry (Incoming Products):** Recording of all incoming product deliveries from suppliers. Inventory levels are updated automatically.

- **Sales Recording (Outgoing Products):** Recording of product sales transactions. Stock quantities are automatically adjusted.
- **Viewing Stock Levels:** Users can check current inventory levels per product and per branch.
- **Generating Reports:** Branch-wise and consolidated inventory reports can be generated for stock monitoring and decision-making.

Operating Hours:

- Monday to Saturday: 8:00 AM - 5:00 PM
- Sunday: 8:00 AM - 12:00 PM

During these hours, all system functionalities are fully available. Outside these hours, users may access the system in view-only mode for monitoring purposes.

Maintenance and Backups:

Routine maintenance, including database backups, will be scheduled during off-peak hours to minimize disruption to operations.

The system includes error handling for network disruptions, allowing cached transactions to synchronize once connectivity is restored.

Special Operations:

Monthly stock audits can be performed using system-generated reports to reconcile physical inventory with digital records.

Low-stock alerts are continuously monitored and logged to assist with timely restocking.

3.1.h Site Adaptation Requirements**Deployment Location:**

The system will be deployed on a web-based hosting environment within the local premises of both Autophoria and Autobox branches. This setup ensures that each branch can securely access and manage the centralized inventory system through their internal network while maintaining reliable connectivity. The deployment allows both branches to operate

efficiently using locally hosted resources, with the option to transition to an online hosting solution in the future if required by the owner.

Pre-Deployment Requirements:

1. Hardware Requirements

- Server computer (for centralized database and system hosting)
- Client computers for employees (per branch)
- Reliable router/switch for network connectivity
- Backup storage device (external drive or NAS)

2. Software Requirements

- Operating System
- Web server software
- Database management system
- Server-side language runtime
- Web browser for client access

3. Network Requirements

- Stable Local Area Network (LAN) within each branch
- Internet connection
- Proper IP addressing and network configuration

4. Communication Protocol Setup

- Configuration of HTTPS for secure communication
- Ensure TCP/IP is properly configured

5. Database Preparation

- Database schema creation
- Initial data setup
- User roles and access control configuration
- Backup and recovery plan

6. Security Requirements

- User authentication system (login/logout)
- Role-based access control (admin, staff, etc.)
- Antivirus and firewall setup on server
- Regular backup configuration

7. Testing Requirements

- Functional testing (inventory updates, search, alerts)
- Network testing between Autophoria and Autobox
- Security testing (login protection, data encryption)
- Performance testing (multiple users accessing system)

8. User Preparation

- Training for employees on how to use the system
- User manuals or documentation
- Admin orientation for system management

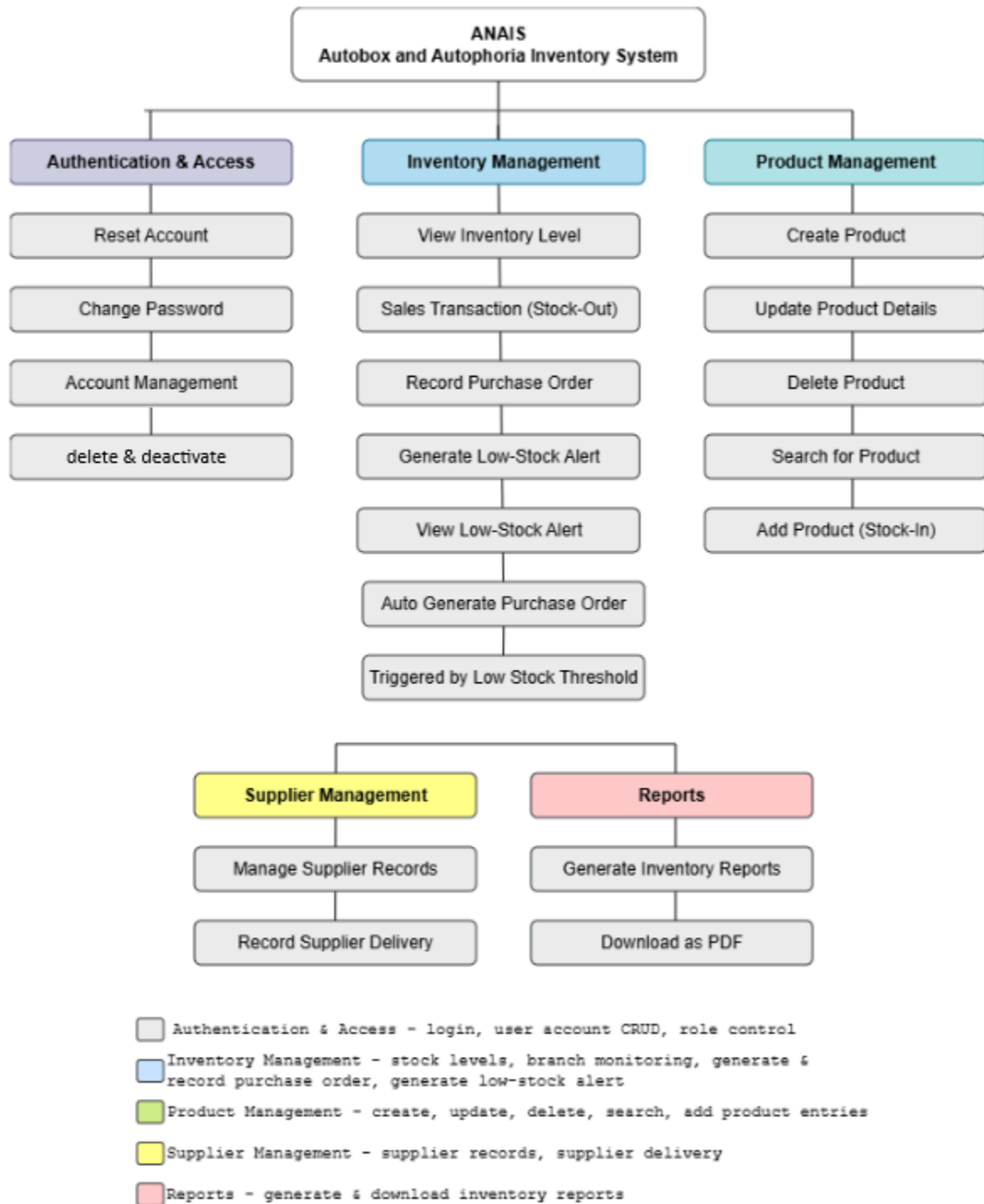
9. Deployment Preparation

- Final system review and debugging
- Data validation and cleanup
- Deployment plan (schedule and steps)
- Rollback plan in case of failure

10. Maintenance Planning

- Schedule for system updates
- Backup schedule (daily/weekly)

3.2 Product Functions



The following table describes all functional capabilities provided by ANAIS, organized by functionality. The system serves three primary user roles – Owner, OIC, and Employee each with defined access levels. The Supplier Management module (Manage

Supplier, Record Purchase Order, Record Supplier Delivery) is included in the current version of the system.

Functionality	Description
Log-In	Allows users (Owner, OIC, Employee) to securely access the system using authorized credentials. The system authenticates users and grants access based on their assigned role.
Log-Out	Allows users (Owner, OIC, Employee) to securely end their session and exit the system. The session is immediately invalidated upon logout and the user is redirected to the login page.
Update User Account Details	Allows the Owner to manage user accounts, including resetting account credentials and updating user details such as username, password, role, and branch assignment. This function ensures secure handling of account-related issues and maintains data accuracy. All account reset and update operations are restricted exclusively to the Owner to enforce role-based access control and prevent unauthorized modifications.
Assign Role (Account Management)	Restricts system features and data visibility depending on the user's assigned role (Owner, OIC, Employee). Each user can only access functions permitted under their role.
Add Product (Stock-In)	Allows the OIC or Owner to record incoming products upon delivery. The system automatically increases the corresponding product's stock quantity in the inventory.
Create New Sales Transaction (Stock-Out)	Allows authorized users to record product sales transactions. The system automatically deducts the sold quantity from the current inventory stock level.
Add / Create Product (Product Management)	Allows the Owner and OIC to add new products to the inventory database, including product name, SKU, category, branch, price, quantity, and low-stock threshold.

Update Product Details	Allows the Owner and OIC to modify existing product details such as name, category, price, or quantity.
Delete Product Details	Allows the Owner to permanently remove a product record from the system. A confirmation prompt is required before deletion is executed.
View Inventory Stock Level	Allows all users to check current stock quantities per product and per branch. Employees have view-only access and cannot modify any data.
Monitor Inventory Across Branches	Allows the Owner and OIC to monitor inventory data of both Autobox La Trinidad and Autophoria Baguio City from a single interface.
Search for Product	Allows users to quickly locate specific products within the inventory using a product name, SKU, or category filter.
Generate Reports (Types of Reports)	Allows the Owner and OIC to generate branch-wise and consolidated inventory reports for stock monitoring and operational decision-making. Reports may be downloaded as PDF files.
Create User Account	Allows the Owner to register a new system user, assigning a specific role (Owner, OIC, Employee) and branch.
View User Account List	Allows the Owner to view details of all registered system accounts, including those of Owners, OICs, and Employees across both Autophoria and Autobox branches. The displayed information includes role assignments and corresponding branch affiliations.
Deactivate User Account	Allows the Owner to deactivate or remove a user account from the system while maintaining audit logs for record-keeping.
Delete User Account	Allows the Owner to permanently remove a user account from the system. All associated access credentials are revoked, and the account is no

	longer retrievable, while maintaining audit logs for record-keeping purposes.
Generate Low-Stock Alerts	The system automatically monitors product quantities and triggers an alert when a product reaches or falls below the predefined minimum stock threshold. Alerts are logged in the system and linked to the relevant product.
View Low-Stock Notification	Allows the Owner and Officer-in-Charge (OIC) to view active low-stock alerts on the dashboard, providing actionable information to restock products promptly.
Add Supplier	Allows the Owner to register new suppliers by entering necessary details such as supplier name, contact information, and associated branch. Suppliers must be registered before they can be used in purchase orders.
View Supplier List	Allows the Owner to view all registered suppliers, including their contact details and current status (active or inactive).
Update Supplier Information	Allows the Owner to modify existing supplier details such as contact information or assigned branch to ensure records remain accurate and up to date.
Activate Supplier	Allows the Owner to set a supplier's status to <i>active</i> , enabling the supplier to be used in purchase orders and other transactions. This function is used when a previously inactive supplier becomes available again for business operations.
Deactivate Supplier	Allows the Owner to set a supplier's status to <i>inactive</i> , preventing the supplier from being selected in future purchase orders while retaining their records for audit and historical reference.
Create Purchase Order	Allows the Owner to create and record a purchase order (PO) for products from a registered supplier. The system assigns a unique PO number and sets the initial status to "pending."

Supplier Delivery	Allows the Owner or OIC to record the actual receipt of products from a supplier against an existing purchase order. Upon confirmation, the system automatically updates inventory stock levels and adjusts the PO status to "partially delivered" or "fully delivered".
Generate Automated Purchase Order (Requires Owner Review & Confirmation)	When a product's quantity reaches or falls below the defined low-stock threshold, the system automatically generates a draft purchase order for the associated supplier. The Owner reviews and confirms the order before it is finalized.
View Purchase Order	Allows the Supplier to view all purchase orders issued by the Owner, including item details, quantities, and order status. The Supplier is automatically notified via email whenever a new purchase order is created or updated, ensuring timely awareness and response.
Confirm Purchase Order	Allows the Supplier to acknowledge and accept a purchase order, indicating that the requested items will be processed for delivery.
Update Delivery Status	Allows the Supplier to update the current status of deliveries (e.g., processing, shipped, in transit, delivered) to keep the system informed in real time.
View Delivery History	Allows the Supplier to access a record of all completed and past deliveries for tracking and reference purposes.
Receive Automated Purchase Order Notifications	Allows the Supplier to receive system-generated notifications whenever a new purchase order is created or updated by the Owner.

3.3 User Characteristics

User	Description	Functions
Owner	Administrator	• User Login • Manage User Access by Role • Create User Account • Update User Account Details • Deactivate User Account • Create Product • Update Product Details • Delete Product • Process Incoming Stock • Process Outgoing Stock (Sales) • View Product List (View Inventory Levels) • Monitor Inventory Across Branches • Search for Product • Generate Reports • Generate Low-Stock Alerts • View Low-Stock Alerts • Add Supplier • View Supplier List • Update Supplier Information • Activate Supplier • Deactivate Supplier • Record Purchase Order • Record Supplier Delivery • Generate Automated Purchase Order

OIC	User	• User Login • Process Incoming Stock • Process Outgoing Stock (Sales) • View Inventory Levels • Monitor Inventory Across Branches • Search for Product • Generate Reports • View Low-Stock Alerts • Record Supplier Delivery
Employee	User	• User Login • View Inventory Levels • Monitor Inventory Across Branches • Search for Product
Supplier	User	• View Purchase Orders • Confirm Purchase Order • Update Delivery Status • View Delivery History • Receive Automated Purchase Order Notifications

3.4 Constraints

The development and implementation of the Autobox and Autophoria Inventory System (AN AIS) are influenced by several organizational, operational, and technical constraints. These limitations shape the system's design, features, and overall functionality.

- **Technical Knowledge** - The developers' current level of technical expertise limits the integration of advanced features such as predictive analytics or automated demand forecasting. As a result, the system will focus on implementing reliable core functionalities such as product listing, stock monitoring, and CRUD (Create, Read, Update, Delete) operations to ensure stability and accuracy.
- **Development Experience** - Limited real-world experience in inventory systems may result in gaps between system design and actual business workflow. This may affect how accurately the system reflects real inventory processes, requiring continuous validation and feedback from the business owner and staff to ensure usability and correctness.
- **Time Constraints** - The system must be completed within the academic project timeline, which restricts the inclusion of complex or non-essential features. Development will prioritize a Minimum Viable Product (MVP), focusing only on core inventory functionalities while deferring advanced features such as analytics dashboards or system integrations.
- **User Technical Skills** - Employees may have limited experience with digital systems, which impacts system design. The interface must be simple, intuitive, and user-friendly, with clear navigation and minimal complexity to reduce errors in data entry and ensure smooth daily operations.
- **Internet Dependency** - Since the system involves centralized data access between Autobox in La Trinidad and Autophoria in Baguio City, unstable internet connectivity may disrupt real-time synchronization. This can lead to delayed updates or inconsistent records between branches, requiring basic data validation or synchronization mechanisms.
- **Hardware Limitations** - The system must operate using existing computers and devices available in both branches. Limited hardware performance may affect system speed and

responsiveness, requiring the system to be lightweight and optimized for low to moderate specifications.

- Scope Limitation - The project is intentionally limited to inventory and stock control functions, including product record management, stock-in and stock-out recording, supplier and purchase order management, low-stock alerts, and report generation. Modules such as accounting, payroll, CRM, and e-commerce are excluded in order to maintain a focused, feasible, and achievable development scope within the academic project timeframe.

3.5 Assumptions and Dependencies

Assumptions

The development and operation of the Autobox and Autophoria Inventory System (AN AIS) are based on the following assumptions:

- Users (owners and employees) have basic knowledge of using web-based systems and can navigate the interface with minimal training.
- All inventory data entered into the system (e.g., stock-in, stock-out, product details) is accurate, complete, and updated regularly by authorized personnel.
- Each branch will follow standardized procedures for recording inventory transactions to ensure consistency across the system.
- The system will be primarily used during business hours with a manageable number of concurrent users.
- The business owner and staff will provide necessary operational information (e.g., product lists, supplier details) during system setup and testing.
- Computers currently available in both branches are functional and capable of running a web-based system.

Dependencies

The functionality and performance of the AN AIS system depend on the following external factors:

- The system depends on a stable internet connection to synchronize data between the two branches.
- The system relies on a centralized database server to store, retrieve, and manage all inventory and business records
- Continuous access to supported web browsers (e.g., Google Chrome, Microsoft Edge) is required for proper system usage and interface rendering.

- The system depends on hosting services to ensure availability, accessibility, and data storage.
- Proper system usage depends on user compliance with proper procedures and data entry practices

3.6 Apportioning of Requirements

The Autobox and Autophoria Inventory System (AN AIS) will be developed with a focus on core operational stability. While the initial release covers the complete end-to-end flow of inventory and procurement, certain high-level integrations and automated optimizations have been scheduled for future iterations to ensure the primary system is robust and bug-free upon deployment.

Requirements Included in the Initial Release (Version 0.5)

- 1. User Management**
 - Role-based authentication and account management for Owner and OIC.
- 2. Core Inventory**
 - Product records, stock-in/stock-out, and barcode/QR scanning.
- 3. Multi-Branch Support**
 - Consolidated dashboards and branch-wise inventory viewing.
- 4. Procurement**
 - Supplier management, automated low-stock alerts, and purchase order tracking.
- 5. Reporting**
 - Basic financial summaries and inventory movement reports.
- 6. Accessibility**
 - Web-based management with mobile interface support for branch staff.

Requirements Deferred for Future Development (Version 0.6 +)

- 1. AI-Driven Demand Forecasting**
 - Implement machine learning algorithms to predict seasonal stock needs based on historical sales data.
- 2. Automated Payroll and Employee Commission Tracking**

- Integrate sales data with a payroll module to automatically calculate staff commissions and performance-based incentives.

3. E-commerce Integration & Customer Portal

- Develop a public-facing interface allowing customers to view real-time stock availability and place pick-up reservations online.

4. Cloud-Based Automated Data Backups and Recovery

- Implement a secondary off-site disaster recovery protocol to ensure 99.9% data availability in case of primary server failure.

5. External Accounting Software Integration (API)

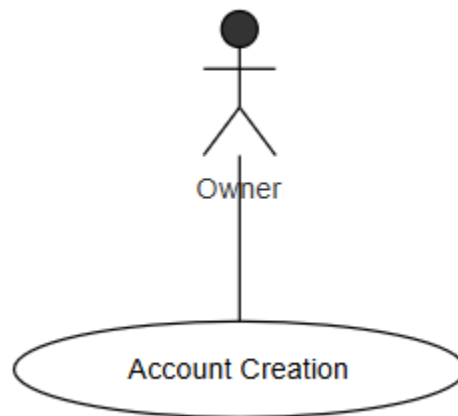
- Seamless data synchronization with third-party software like QuickBooks or Xero for professional-grade tax and audit preparation.

4 Specific Requirements

4.1 Functional Requirements

This section details the functional requirements of ANAIS through structured use cases. Each use case describes a specific interaction between users and the system, including basic flows, alternate flows, exception flows, and related business rules.

4.1.1 ANAIS-UC01 (Account Creation)



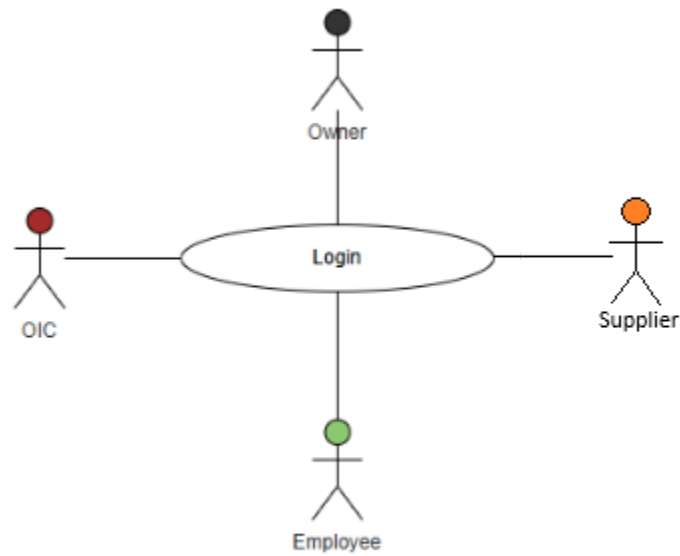
ANAIS - UC01	Account Creation
Description	This use case allows the Owner to create accounts for OICs and Employees to access the system.
Actor	Owner (Administrator)
Trigger	The Owner needs to create an account for branch staff (OIC or Employee) to access ANAIS.

Complexity	Easy	
Pre-condition	• Owner must be logged in to the system. • Owner is logged into the system and the Account Management page is currently open. • Account creation form is displayed.	
Post condition	• New user account is successfully saved and ready to use.	
Basic flow	Actor Action	System Response
	Step 1. Enters employee account details (name, role, branch, username, password) and submits the form.	Step 2. The system verifies and validates the submitted details. Step 3. The system provides an alert confirming that the account has been created. Step 4. The system notifies the Owner that account creation is complete and the account is ready to use.
Alternate Flow	Actor Action	System Response
	Step 2a. Invalid or incomplete details entered.	Step 3a. System detects an existing account with the same details.

	Step 2a.1 System notifies the user to enter valid details. Step 2a.2 Returns to Step 1.	Step 3a.1 - System displays an error message indicating that the username is already taken and prompts the user to choose a different username. Step 3a.2 Returns to Step 1.
Exception Flow	When the system experiences a critical error or session failure: • System will terminate the current session, log out the Owner, and redirect them to the login page for re-authentication.	
Includes	AN AIS-UC02 (Login), AN AIS-UC03 (Logout)	
Extends	AN AIS-UC02 (Login), AN AIS-UC03 (Logout)	
Assumptions		
Business Rules	• Only the Owner (Administrator) has the privilege to create accounts. • Accounts can be assigned one of three roles: Owner, OIC, or Employee. • Each username must be unique across the system. • Only the Owner can create user accounts.	
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.1 Account Creation Form	
Field Validations	5.2.1 Data Description for Account Management	

Author	Albino, Erin Camille; Zapanta, Zeus Levi
Date	March 2026

4.1.2 ANAIS-UC02 (Account Login)



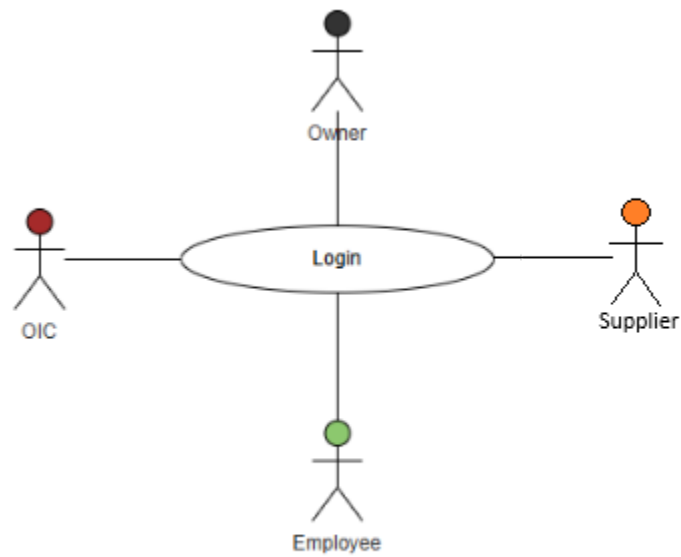
ANAIS - UC01	Account Login
Description	This use case allows authorized users (Owner, OIC, Employee, Supplier) to access the ANAIS system using their credentials.
Actor	Owner, OIC, Employee, Supplier

Trigger	A user needs to log in to access system features.	
Complexity	Easy	
Pre-condition	Login form must be displayed on the landing page.	
Post condition	The user is successfully authenticated and redirected to the dashboard based on their role.	
Basic flow	Actor Action	System Response
	Step 1. User enters their login credentials on the login form and submits.	Step 2. System validates and verifies the login credentials. Step 3. System redirects the user to the dashboard appropriate to their role (Owner, OIC, or Employee)
Alternate Flow	Actor Action	System Response
	Step 2a. Invalid or incomplete credentials entered, or the account does not exist in the system.	Step 3a. Login credentials do not match any record in the database. Step 3a.1 System prompts the user that the username or password is incorrect.

	Step 2a.1 System notifies user to enter valid login details or informs them that the account is not registered. Step 2a.2 Returns to Step 1.	Step 3a.2 Returns to Step 1 (Login page) to allow the user to re-enter correct credentials.
Exception Flow	When a system error occurs during authentication (such as server failure, database connection failure, or session creation error): the login process will be terminated and the user will be redirected to the login page.	
Includes	AN AIS-UC01 (Account Creation)	
Extends	AN AIS-UC03 (Logout)	
Assumptions	User has an existing registered account in the system.	
Business Rules	- Only users who are authorized to the system can log in. - Access level is determined by the role assigned during account creation. - The system grants view-only access to Employees and limited data-entry rights to OICs.	
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.2 Login Form	
Field Validations	5.2.1 Data Description for Account Management	

Author	Arjona, Clifford Xhyrus; Lopez, Hans Jayden; Albino, Erin Camille
Date	March 2026

4.1.3 ANAIS-UC03 (Logout)

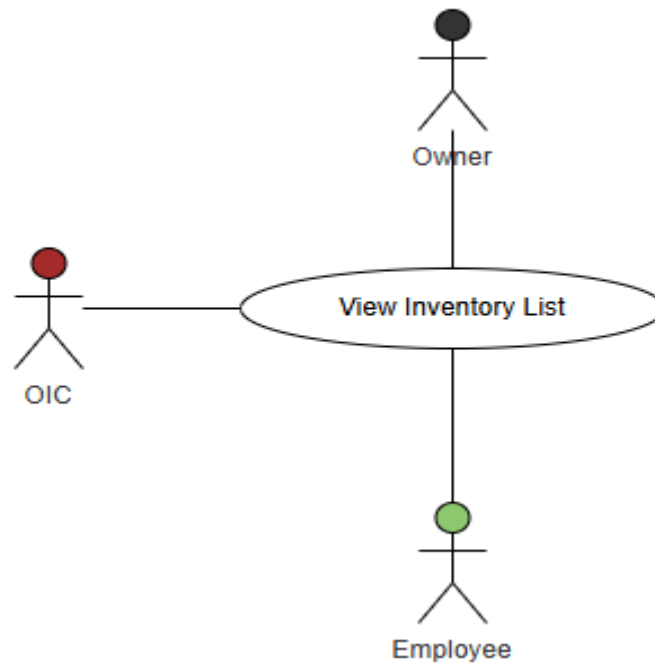


ANAIS - UC03	Account Logout
Description	This use case allows users to securely log out of the ANAIS system.
Actor	Owner, OIC, Employee, Supplier

Trigger	The user wants to end their current session and log out of the system.	
Complexity	Easy	
Pre-condition	The user must be authenticated and have an active session in the system.	
Post condition	The user is successfully logged out and redirected to the landing/login page.	
Basic flow	Actor Action	System Response
	Step 1. User clicks the Logout button from the navigation menu.	Step 2. System ends the active session and redirects the user to the login/landing page.
Alternate Flow	Actor Action	System Response
Exception Flow	If the system fails, the user will be automatically logged out and redirected to the login page.	
Includes	AN AIS-UC02 (Login)	
Extends		

Assumptions	
Business Rules	• Any data entered but not saved prior to logout will be discarded. • Session tokens are invalidated immediately upon logout.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.3 Logout
Field Validations	5.2.1 Data Description for Account Management
Author	Cauan, Nel Angelo; Albino, Erin Camille
Date	March 2026

4.1.4 ANAIS-UC04 (View Inventory List)

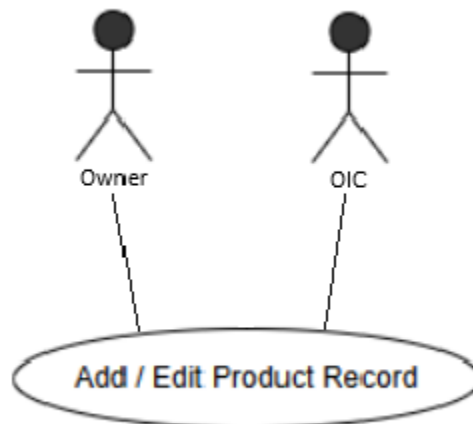


ANAIS - UC04	View Inventory List
Description	This use case allows authorized users to view current inventory levels and product details across branches.
Actor	Owner, OIC, Employee
Trigger	The user wants to view the current stock levels or products list information for a branch.
Complexity	Easy
Pre-condition	The user must be logged in to the system.

	The inventory list page must be accessible from the dashboard.	
Post condition	The user is able to view the inventory list with current stock quantities per product and per branch.	
Basic flow	Actor Action	System Response
	Step 1. User chooses to view a branch inventory.	Step 2. System retrieves and displays the inventory list with referenced data.
Alternate Flow	Actor Action	System Response
	Step 2a. No products found in the selected branch.	Step 2a.1 System displays a message indicating no records available.
Exception Flow	When the system fails while displaying the inventory list: An error message will be shown and the user will be redirected to the login page to re-establish the session.	
Includes	ANAIS-UC02 (Login)	
Extends	Search Filter: It is triggered when the user enters a search keyword or selects filter criteria.	

Assumptions	
Business Rules	• All user roles (Owner, OIC, Employee) can view inventory. • Employees have view-only access and cannot edit any inventory data. • The system displays real-time inventory data from the central database.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.4 Inventory View
Field Validations	5.2.2 Data Description for Inventory Records
Author	Lopez, Hans Jayden; Arjona, Clifford Xhyrus
Date	March 2026

4.1.5 ANAIS-UC05 (Add/Edit Product Record)

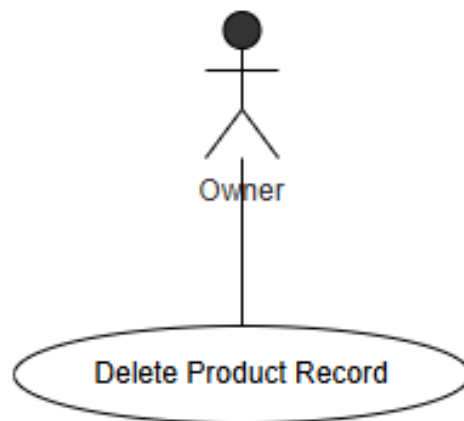


AN AIS - UC05	Add / Edit Product Record	
Description	This use case allows the Owner and OIC to add new products or edit existing product records in the inventory.	
Actor	Owner, OIC	
Trigger	The Owner needs to add a new product or update the details of an existing product.	
Complexity	Easy	
Pre-condition	• The Owner must be logged in to the system. • The product management form must be displayed.	
Post condition	• The product record is successfully added or updated in the inventory database.	
Basic flow	Actor Action	System Response
	Step 1. Owner enters or updates product details.	Step 2. System verifies the submitted details for completeness and validity. Step 3.

		System validates the database for duplicate SKU entries. Step 4. System confirms and notifies the Owner that the product record has been saved successfully.
Alternate Flow	Actor Action	System Response
	Step 2a. Submitted input is invalid or incomplete. Step 3a. Duplicate SKU detected in the database.	Step 2a.1 System displays field-level validation errors. Step 2a.2 Returns to Step 1. Step 3a.1 System prompts Owner to use a unique SKU. Step 3a.2 Returns to Step 1.
Exception Flow	• If the system fails, the Owner will be logged out and redirected to the login page. • Unsaved data will be discarded.	
Includes	ANAIS-UC02 (Login), ANAIS-UC04 (View Inventory)	
Extends		

Assumptions	• The system database is assumed to be operational, accessible, and properly connected at the time of transaction. • It is assumed that all required product information is provided in valid and acceptable formats before submission. • Each product is assumed to have a unique identifier to prevent duplication of records within the database. • A stable network connection is assumed to be available during the execution of add or edit operations to ensure successful data transmission and storage. • It is assumed that any updates made to product records are immediately reflected in the centralized database and are accessible across both branches of the system.
Business Rules	• The Owner and OIC are authorized to add and edit product records. • Each product must be assigned to a specific branch (Autobox or Autophoria). • SKU must be unique per product.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.5 Product Management Form
Field Validations	5.2.2 Data Description for Inventory Records
Author	Zapanta, Zeus Levi; Albino, Erin Camille
Date	March 2026

4.1.6 ANAIS-UC06 (Hard Delete Product Record)

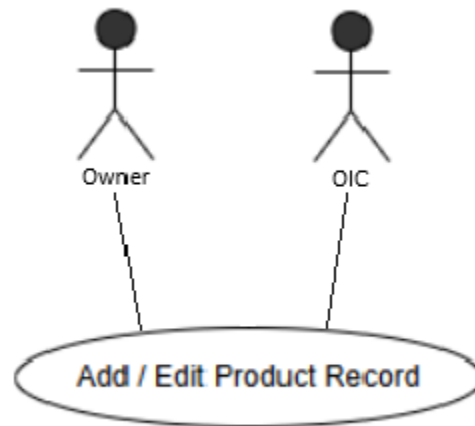


ANAIS - UC06	Hard Delete Product Record
Description	This use case allows the Owner to remove a product record from the inventory system.
Actor	Owner
Trigger	Owner needs to remove a product that is no longer carried or relevant.
Complexity	Easy
Pre-condition	• The Owner must be logged in to the system. • The product to be deleted must exist in the inventory.

Post condition	The selected product record is permanently removed from the system.	
Basic flow	Actor Action	System Response
	Step 1. Owner selects a specific product and chooses the Delete option. Step 3. Owner confirms the deletion.	Step 2. System displays a confirmation to confirm the deletion. Step 4. System removes the product and notifies the Owner that the deletion was successful.
Alternate Flow	Actor Action	System Response
	Step 3a. Owner chooses to cancel the deletion.	Step 3a.1 System returns to the product list without making changes.
Exception Flow	If an error occurs during the deletion process, such as a database connection failure, server error, or network interruption: - The Owner will be logged out and redirected to the login page. - The system shall display an error message indicating that the deletion was unsuccessful.	

	• The product record shall remain unchanged in the system.
Includes	AN AIS-UC02 (Login), AN AIS-UC04 (View Inventory)
Extends	
Assumptions	• The user performing the stock-in operation is an authorized role (Owner or OIC) with permission to record incoming inventory transactions. • The system is assumed to exclude deleted products from the default inventory list view unless explicitly requested.
Business Rules	• Only the Owner can delete product records. • Deletion is permanent and cannot be undone.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.6 Delete Confirmation
Field Validations	5.2.2 Data Description for Inventory Records
Author	Cauan, Nel Angelo; Zapanta, Zeus Levi
Date	March 2026

4.1.7 ANAIS-UC07 (Update Inventory Stock-In)



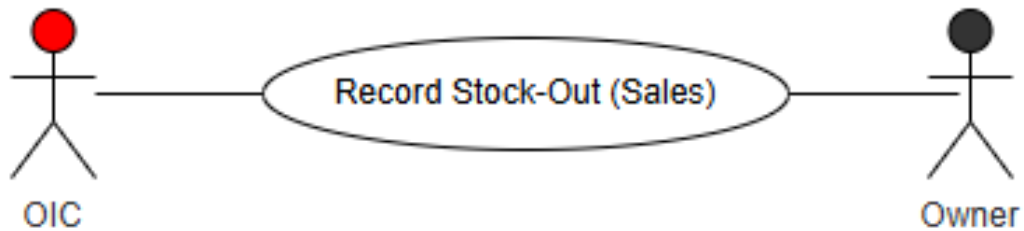
ANAIS - UC07	Update Inventory Stock-In
Description	This use case allows the Owner and OIC to record incoming products (deliveries) and update inventory stock levels.
Actor	OIC (Officer-in-Charge)
Trigger	New products are delivered to the branch and must be recorded in the system.
Complexity	Easy
Pre-condition	• The OIC must be logged in to the system. • The stock-in form must be displayed.
Post-condition	• The incoming product quantities are added to the inventory and the stock levels are updated in real time.

Basic flow	Actor Action	System Response
	Step 1. OIC selects the product from the inventory list and enters the received quantity and delivery date.	Step 2. System ask for a confirmation to verify the updated inventory list submitted. Step 3. System confirms the stock-in transaction and notifies the OIC. Step 4. System updates the inventory by adding the received quantity to the existing stock.
Alternate Flow	Actor Action	System Response
	Step 2a. Invalid or incomplete details submitted.	Step 2a.1 System prompts OIC to correct the fields. Step 2a.2 Returns to Step 1.
Exception Flow	• If the quantity entered is invalid, the system shall display an error message and require valid input.	

	• If duplicate requests are detected, the system shall process only one and prevent duplication.
Includes	AN AIS-UC02 (Login), AN AIS-UC04 (View Inventory)
Extends	• Record Supplier Delivery • Generate Low-Stock Update
Assumptions	• The user has permission to perform stock-in operations. • The product to be updated already exists and is active. • Stock-in updates are immediately reflected in the centralized inventory across both branches.
Business Rules	• Only the OIC and Owner can record stock-in transactions. • The stock quantity is automatically incremented upon confirmation. • Each stock-in transaction is logged with a timestamp and the name of the user who performed it.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.7 Stock-In Form
Field Validations	5.2.3 Data Description for Stock Transactions
Author	Arjona, Clifford Xhyrus; Lopez, Hans Jayden; Albino, Erin Camille

Date	March 2026
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4.1.8 ANAIS-UC08 (Record Stock-Out (Sales))



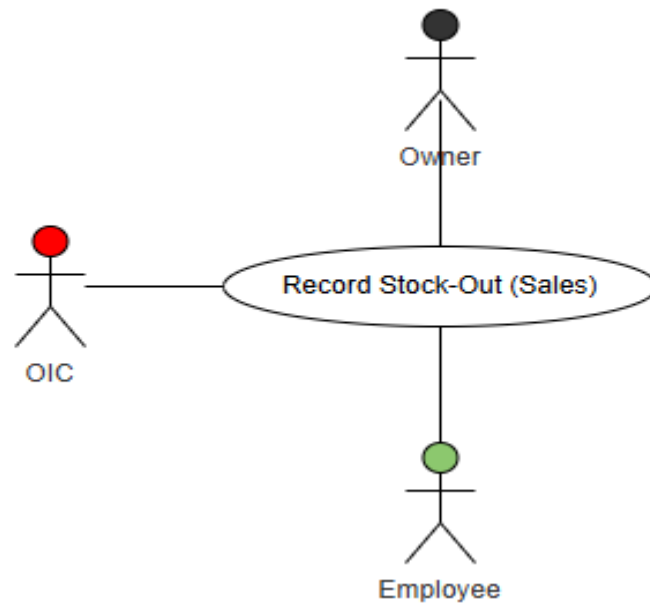
ANAIS - UC08	Record Stock-Out (Sales)
Description	This use case allows authorized users to record product sales, automatically deducting quantities from the inventory.
Actor	Owner, OIC
Trigger	A product is sold and must be deducted from the inventory.
Complexity	Easy

Pre-condition	- The user must be logged in to the system. - The product to be sold must exist in the inventory with sufficient quantity.	
Post condition	The sold product quantity is deducted from the inventory and the stock level is updated.	
Basic flow	Actor Action	System Response
	Step 1. User selects the product sold and enters the quantity sold.	Step 2. System verifies the submitted details and checks for sufficient stock. Step 3. System deducts the sold quantity from the current inventory. Step 4. System confirms the sale transaction.
Alternate Flow	Actor Action	System Response

	Step 2a. Insufficient stock available.	Step 2a.1 System notifies the user that the quantity requested exceeds available stock. Step 2a.2 Returns to Step 1.
Exception Flow	• If the system fails, the user will be logged out and redirected to the login page. • Unsaved transactions will be discarded.	
Includes	ANAIIS-UC02 (Login), ANAIIS-UC04 (View Inventory)	
Extends	• ANAIIS-UC10 (Receive Low-Stock Alert) • Generate Low-Stock Alert	
Assumptions	• The user performing the stock-out transaction is an authorized role (Owner or OIC) with permission to record sales transactions in the system. • The product being sold exists in the system and is active. • The product has sufficient available stock to fulfill the requested quantity. • The stock-out transaction immediately updates the inventory levels in the centralized database across both branches.	

	• The system maintains records of all sales transactions for reporting and auditing purposes.
Business Rules	• Only the Owner and OIC can record sales transactions. • The system prevents recording a sale if the product has zero stock. • A low-stock alert is triggered if the remaining quantity falls below the defined threshold.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.8 Stock-Out Form
Field Validations	5.2.3 Data Description for Stock Transactions
Author	Albino, Erin Camille; Cauan, Nel Angelo
Date	March 2026

4.1.9 ANAIS-UC09 (Search for Product)

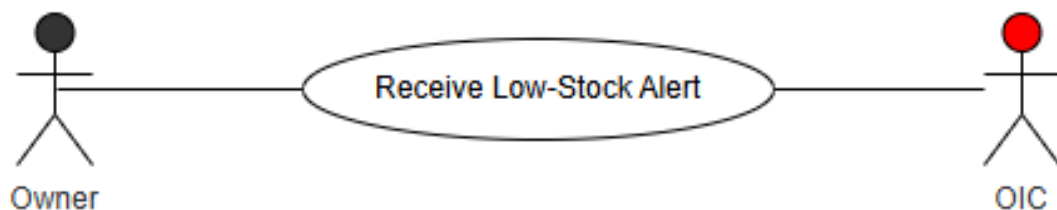


ANAIS - UC09	Search for Product
Description	This use case allows users to quickly locate specific products in the system, like products available using search criteria.
Actor	Owner, OIC, Employee
Trigger	The user needs to find a specific product by name or category.
Complexity	Easy

Pre-condition	- The user must be logged in to the system. - Products must be encoded in the inventory database.	
Post condition	The system displays a filtered list of products matching the search criteria.	
Basic flow	Actor Action	System Response
	Step 1. User enters a product name or selects a category in the search field.	Step 2. System retrieves and displays products matching the search criteria.
Alternate Flow	Actor Action	System Response
	Step 2a. No products match the search criteria.	Step 2a.1 System displays a message: 'No products found.'
Exception Flow	If the system fails, the user will be logged out and redirected to the login page.	
Includes	AN AIS-UC02 (Login)	
Extends	Filter Search Results: when the user selects filter criteria. The system refines and displays results based on selected filters.	

	View Product Details: When the user selects a product from the results. The system displays detailed product information.
Assumptions	- Product records already exist in the system. - The user enters valid search input.
Business Rules	- All user roles can perform product searches. - Search results are filtered by the branch the user belongs to, unless the user is the Owner who can view all branches.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.9 Search Product
Field Validations	5.2.2 Data Description for Inventory Records
Author	Lopez, Hans Jayden; Arjona, Clifford Xhyrus
Date	March 2026

4.1.10 ANAIS-UC10 (Receive Low-Stock Alert)



ANAIS - UC10	Receive Low-Stock Alert
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Description	This use case notifies authorized users when a product's quantity falls at or below the defined minimum stock threshold.	
Actor	Owner, OIC	
Trigger	A product's stock quantity reaches or falls below the predefined reorder threshold.	
Complexity	Easy	
Pre-condition	- The user must be logged in. - Products must have a defined low-stock threshold.	
Post condition	The Owner and OIC receive a low-stock notification visible on the dashboard.	
Basic flow	Actor Action	System Response
	Step 4. The Owner and OIC receive a low-stock alert	Step 1. The system automatically checks inventory levels after each stock-out transaction. Step 2. System detects that a product's quantity is at or below the low-stock threshold.

		Step 3. The system generates and displays a low-stock alert on the dashboard for the Owner and OIC.
Exception Flow	If the system fails, the alert may not be displayed until the system is restored.	
Includes	ANAIS-UC08 (Record Stock-Out)	
Extends	ANAIS-UC16 (Automated Purchase Order Generation)	
Assumptions	- The system continuously monitors inventory levels in real time. - A predefined low-stock threshold is configured for each product. - Product stock levels are accurately updated through stock-in and stock-out transactions. - The user has permission to view inventory notifications and alerts.	
Business Rules	- Only the Owner and OIC receive low-stock alerts. - The low-stock threshold is configured per product by the Owner. - Alerts remain visible until the stock is replenished above the threshold.	
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.10 Low-Stock Alert Display	

Field Validations	5.2.2 Data Description for Inventory Records
Author	Zapanta, Zeus Levi; Albino, Erin Camille
Date	March 2026

4.1.11 ANAIS-UC11 (Generate Reports)



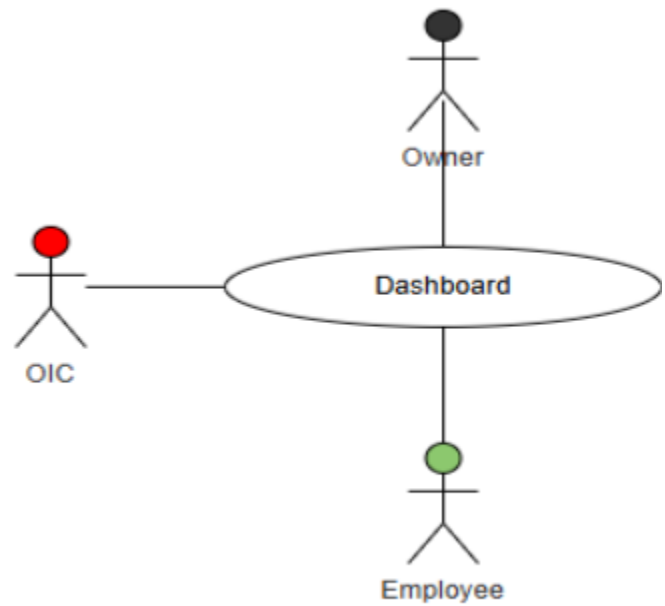
ANAIS - UC11	Generate Reports
Description	This use case allows the Owner and OIC to generate branch-wise and consolidated inventory reports.
Actor	Owner, OIC
Trigger	The Owner or OIC needs to generate an inventory report for monitoring or decision-making.
Complexity	Easy

Pre-condition	The user must be logged in to the system.	
Post condition	The system generates and displays the requested inventory report.	
Basic flow	Actor Action	System Response
	Step 1. User sets and submits the report filter criteria (branch, date range, product category).	Step 2. System retrieves and matches records based on the filter criteria. Step 3. System generates and displays the report. User may download or print the report.
Alternate Flow	Actor Action	System Response
	Step 2a. No records match the filter criteria.	Step 2a.1 System displays a message indicating no data found for the selected filters.
Exception Flow	If the system fails, the user will be logged out and redirected to the login page.	
Includes	ANAIS-UC02 (Login), ANAIS-UC04 (View Inventory)	

Extends	● Filter Report Data: When the user selects filters, the system generates a report based on the selected criteria. ● Export Report: When the user chooses to export the report, the system allows the report to be downloaded in formats such as PDF or Excel. ● View Report Summary: After report generation, the system displays a summarized version of the report before export or printing.
Assumptions	● The user generating the report is an authorized role (Owner or OIC) with permission to access and generate inventory reports. ● Relevant data is available and up to date in the database. ● Report templates are predefined in the system. ● The system contains accurate and complete transaction logs for reporting purposes.
Business Rules	● Only the Owner and OIC can generate reports. ● Reports can be generated per branch or as a consolidated multi-branch report. ● The generated report may be downloaded as a PDF file.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.11 Generate Reports
Field Validations	5.2.4 Data Description for Report Generation
Author	Albino, Erin Camille; Cauan, Nel Angelo

Date	March 2026
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4.1.12 ANAIS-UC12 (Dashboard)



ANAIS - UC12	Dashboard
Description	This use case allows users to view the ANAIS dashboard, which displays a summary of inventory status and alerts.
Actor	Owner, OIC, Employee
Trigger	The user logs in and is redirected to the dashboard, or navigates to it from the menu.

Complexity	Easy	
Pre-condition	The user must be logged in to the system.	
Post condition	The user can view the dashboard with current stock summaries, branch status, and low-stock alerts.	
Basic flow	Actor Action	System Response
		Step 1. The system automatically displays the dashboard upon successful login Step 2. Dashboard shows stock summary, branch overview, and any active low-stock alerts based on the user's role and branch access.
Alternate Flow	Actor Action	System Response
	Step 1. User with Limited Role Access Step 2. System Delay or Loading Issue	Step 1a. The system displays a simplified dashboard showing only allowed data such as assigned branch inventory and hides administrative modules.

		Step 2a. The system displays a loading indicator and retrieves data once the connection or processing is restored.
Includes	AN AIS-UC02 (Login)	
Extends	• View Detailed Low-Stock Alerts • Filter Dashboard Data • Generate Quick Reports • Navigate to Inventory Module	
Assumptions	• The user role (Owner, OIC, Employee) determines what data is visible on the dashboard. • Inventory, sales, and alert data are already stored and updated in the database. • Low-stock thresholds are already configured in the system. • Data shown in the dashboard is assumed to be up to date at the time of login. • The system automatically loads the dashboard after successful authentication.	
Business Rules	• Dashboard content is role-based: the Owner sees consolidated data for both branches; OIC sees their assigned branch data; Employees see limited view-only data. • Low-stock alerts are prominently displayed on the Owner's and OIC's dashboards.	

Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.12 Dashboard 5.1.13 Landing Page
Field Validations	5.2.4 Data Description for Report Generation
Author	Lopez, Hans Jayden; Arjona, Clifford Xhyrus; Albino, Erin Camille
Date	March 2026

4.1.13 ANAIS-UC13 (Manage Supplier)



ANAIS - UC13	Manage Supplier
Description	This use case allows the Owner to add, edit, or deactivate supplier records in the ANAIS system. Suppliers are linked to products and used when recording purchase orders and deliveries.
Actor	Owner (Administrator)

Trigger	The Owner needs to register a new supplier or update an existing supplier's information.	
Complexity	Easy	
Pre-condition	• The Owner must be logged in to the system. • The Supplier Management form must be displayed.	
Post condition	• The supplier record is successfully saved or updated in the system database.	
Basic flow	Actor Action	System Response
	Step 1. Owner navigates to the Supplier Management section and selects Add Supplier or selects an existing supplier to edit. Step 2. Owner enters or updates the supplier details and submits.	Step 3 System verifies the submitted details for completeness and validity. Step 4. The system checks the database for duplicate supplier names. Step 5. The system saves the supplier record and notifies the Owner that the operation was successful.

Alternate Flow	Actor Action	System Response
	Step 3a. Invalid or incomplete details submitted.	Step 3a.1 System displays field-level validation errors. Step 3a.2 Returns to Step 2. Step 4a. Duplicate supplier name detected. Step 4a.1 System prompts the Owner to use a different supplier name. Step 4a.2 Returns to Step 2.
Exception Flow	• If the system fails, the Owner will be logged out and redirected to the login page. • Unsaved data will be discarded.	
Includes	ANAIS-UC02 (Login)	
Extends	• Search Supplier • View Supplier Details • Activate/Deactivate Supplier • Link Supplier to Purchase Order	

Assumptions	• The user has sufficient privileges to manage supplier records. • Supplier data is entered in valid format. • Each supplier record has a unique identifier in the system. • Supplier records are used in related modules such as purchase orders and deliveries. • Changes made to supplier information are immediately reflected in the centralized database.
Business Rules	• Only the Owner can add, edit, or deactivate supplier records. • Supplier names must be unique in the system. • Deactivated suppliers cannot be selected in new purchase orders but historical records are retained. • Each supplier may be linked to one or more products.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.14 Supplier Management Form
Field Validations	5.2.5 Data Description for Supplier Records
Author	Zapanta, Zeus Levi; Albino, Erin Camille
Date	March 2026

4.1.14 ANAIS-UC14 (Record Purchase Order)



ANAIS - UC14	Record Purchase Order
Description	This use case allows the Owner to create and record a purchase order (PO) for products from a registered supplier. A PO documents the intent to purchase and serves as the basis for tracking incoming deliveries.
Actor	Owner (Administrator)
Trigger	The Owner needs to place an order for products from a supplier to replenish stock.
Complexity	Easy
Pre-condition	• The Owner must be logged in to the system. • At least one active supplier must be registered in the system. • The products to be ordered must exist in the inventory.

Post condition	The purchase order is recorded in the system with a status of 'Pending' and is ready for delivery tracking.	
Basic flow	Actor Action	System Response
	Step 1. The owner navigates to the Purchase Orders section and selects Create New Purchase Order. Step 2. The owner selects the supplier, branch, and the products to order, specifying quantity and unit cost for each item. Step 3. The owner sets the expected delivery date and adds optional	Step 4. The system verifies all submitted details for completeness and validity. Step 5. The system saves the purchase order with a status of 'Pending' and assigns a unique PO number. Step 6. The system notifies the Owner that the purchase order has been successfully recorded.

	remarks, then submits the PO.	
Alternate Flow	Actor Action	System Response
	Step 4a. Invalid, incomplete, or zero-quantity order details submitted.	Step 4a.1 System displays field-level validation errors. Step 4a.2 Returns to Step 2.
Exception Flow	• If the system fails, the Owner will be logged out and redirected to the login page. • Unsaved purchase orders will be discarded.	
Includes	AN AIS-UC02 (Login), AN AIS-UC13 (Manage Supplier)	
Extends	AN AIS-UC15 (Record Supplier Delivery)	
Assumptions	• The user is assumed to have the appropriate access rights and system permissions required to perform this operation. • Supplier records already exist and are active in the system. • The products included in the purchase order already exist in the inventory database. • All required purchase order details are entered in valid format.	

	• Purchase order records are stored and accessible for tracking and reporting purposes. • The system supports linking purchase orders to suppliers and future delivery records.
Business Rules	• Only the owner can create purchase orders. • Each PO must be linked to exactly one active supplier. • A PO may contain one or more product line items. • Each PO is assigned a unique PO number by the system upon saving. • A PO has one of the following statuses: Pending, Partially Delivered, Fully Delivered, or Cancelled. • A PO cannot be edited once it has an associated delivery record; it may only be cancelled.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.15 Purchase Order Form
Field Validations	5.2.6 Data Description for Purchase Orders
Author	Cauan, Nel Angelo; Zapanta, Zeus Levi
Date	March 2026

4.1.15 ANAIS-UC15 (Record Supplier Delivery)



ANAIS - UC15	Record Supplier Delivery
Description	This use case allows the Owner or OIC to record the actual delivery of products from a supplier against an existing purchase order. Upon confirmation, the system automatically updates the inventory stock levels for the delivered items.
Actor	Owner, OIC
Trigger	Products ordered from a supplier have arrived at the branch and must be recorded as received.
Complexity	Easy
Pre-condition	• The user must be logged in to the system. • A purchase order with a status of 'Pending' or 'Partially Delivered' must exist for the supplier. • The delivery must correspond to products listed in the referenced purchase order.

Post condition	• The delivery is recorded in the system. • Inventory stock levels for delivered products are automatically updated (stock-in). • The referenced purchase order status is updated to 'Partially Delivered' or 'Fully Delivered' accordingly.	
Basic flow	Actor Action	System Response
	Step 1. The user navigates to the Supplier Deliveries section and selects Record New Delivery. Step 2. The user selects the purchase order reference (PO number) from the list of pending orders. Step 4. The user enters the actual quantity received for each product line item and the delivery	Step 3. The system displays the expected products and quantities from the selected PO. Step 5. The system verifies the submitted details and checks that received quantities do not exceed the ordered quantities. Step 6. The system records the delivery, updates inventory stock levels by adding the received quantities, and updates the PO status. Step 7. The system notifies the user that the delivery has been successfully recorded

	date, then submits.	and inventory has been updated.
Alternate Flow	Actor Action	System Response
	Step 5a. Received quantity exceeds the ordered quantity for a line item. Step 5b. Invalid or incomplete details submitted	Step 5a.1 System notifies the user that the quantity entered exceeds the ordered amount. Step 5a.2 Returns to Step 4. Step 5b.1 System displays field-level validation errors. Step 5b.2 Returns to Step 4.
Exception Flow	• If the system fails, the user will be logged out and redirected to the login page. • Unsaved delivery records will be discarded and inventory will not be updated.	
Includes	ANAIS-UC02 (Login), ANAIS-UC14 (Record Purchase Order)	
Extends	ANAIS-UC07 (Record Stock-In)	

Assumptions	• The user is assumed to have the appropriate access rights and system permissions required to perform this operation. • A corresponding purchase order already exists in the system for the delivery being recorded. • Supplier records are already registered and active in the system. • Delivered products exist in the inventory database and match the purchase order details. • All delivery details are entered in valid format. • Delivery records are stored for tracking, auditing, and inventory update purposes. • Recorded deliveries automatically update stock-in quantities in the centralized inventory system.
Business Rules	• Only the Owner and OIC can record supplier deliveries. • A delivery must always reference an existing purchase order. • The received quantity per line item cannot exceed the ordered quantity on the PO. • Partial deliveries are allowed; the PO status will be set to 'Partially Delivered' until all items are received. • Once all items on a PO are fully received, the PO status is automatically changed to 'Fully Delivered'. • Each delivery transaction is logged with a timestamp and the name of the user who recorded it.

	Inventory stock levels are automatically incremented upon delivery confirmation.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.16 Supplier Delivery Form
Field Validations	5.2.6 Data Description for Purchase Orders 5.2.7 Data Description for Supplier Deliveries
Author	Albino, Erin Camille; Lopez, Hans Jayden
Date	March 2026

4.1.16 ANAIS-UC16 (Automated Purchase Order Generation)

ANAIS - UC16	Automated Purchase Order Generation
Description	This use case allows the system to automatically generate a draft purchase order when a product's stock quantity reaches or falls below its defined low-stock threshold and the product has a registered supplier. The system creates the draft PO and notifies the Owner for review. The Owner must confirm or cancel the draft before it becomes an active purchase order.
Actor	System (automated), Owner

Trigger	A product's stock quantity reaches or falls below the defined low-stock threshold following a stock-out or supplier delivery transaction, and the product has an active registered supplier.	
Complexity	Easy	
Pre-condition	• The product must have a defined low-stock threshold. • The product must be linked to at least one active registered supplier. • The Owner must be logged in to review and confirm the draft PO. • A low-stock alert (UC10) must have been triggered for the product.	
Post condition	• A draft purchase order is generated by the system and made available for Owner review. • The Owner is notified of the pending draft PO via the dashboard alert panel. • Upon Owner confirmation, the draft PO status is changed to Pending and becomes an active purchase order. • If the Owner cancels, the draft PO is discarded and no purchase order is created.	
Basic flow	Actor Action	System Response
	Step 5. The owner navigates to	Step 1. The system detects that a product's quantity has

	the Automated PO Review screen (5.1.17) and reviews the draft PO details. Step 6. The owner adjusts the order quantity or other details if necessary, then confirms the draft PO.	reached or fallen below the low-stock threshold following a transaction. Step 2. The system checks if the product has an active registered supplier linked to it. Step 3. The system automatically generates a draft purchase order containing the product, its associated supplier, the branch, and a suggested reorder quantity based on the low-stock threshold. Step 4. The system displays a notification on the Owner's dashboard indicating that a draft PO has been generated and is awaiting review. Step 7. The system saves the confirmed PO with a status of Pending and assigns a unique PO number. Step 8.
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		The system notifies the Owner that the purchase order has been successfully created.
Alternate Flow	Actor Action	System Response
	Step 2a. No active supplier is linked to the product. Step 6a. Owner cancels the draft PO review.	Step 2a.1 System generates a low-stock alert only (UC10) and does not create a draft PO. Step 2a.2 System displays a notification advising the Owner to assign a supplier to the product. Step 6a.1 System discards the draft PO. Step 6a.2 The low-stock alert remains visible on the dashboard until stock is replenished.

Exception Flow	• If the system fails during automated PO generation, no draft PO is created. The low-stock alert will remain visible and the Owner may manually create a purchase order via UC14. • If the Owner is not logged in at the time of generation, the draft PO notification will be displayed upon the Owner's next login.
Includes	ANAIS-UC10 (Receive Low-Stock Alert), ANAIS-UC14 (Record Purchase Order)
Extends	• Generate Purchase Order Based on Low Stock: When a product reaches or falls below the predefined minimum stock level the system automatically creates a draft purchase order for the affected product. • Review Generated Purchase Order: When the Owner or OIC chooses to verify the auto-generated order, the system displays the generated purchase order for confirmation or editing. • Edit Auto-Generated Purchase Order: When modifications are required before approval, the system allows changes to quantity, supplier, or product details.
Assumptions	• The product's low-stock threshold has been correctly configured by the Owner. • The product is linked to an active supplier in the system. • The Owner will review and action draft POs in a timely manner.

Business Rules	• Automated PO generation is triggered only when a product has an active registered supplier. If no supplier is linked, only a low-stock alert is raised. • The system generates one draft PO per low-stock event per product. Duplicate draft POs for the same product are not created if a draft is already pending review. • The suggested reorder quantity in the draft PO is calculated as twice the low-stock threshold by default. The Owner may adjust this before confirming. • A draft PO has a status of Draft and is not considered an active purchase order until confirmed by the Owner. • Only the Owner can review, confirm, or cancel draft purchase orders. • Confirmed draft POs follow the same lifecycle as manually created POs (UC14): Pending, Partially Delivered, Fully Delivered, or Cancelled. • The system retains a log of all automated PO generation events, including whether the draft was confirmed or cancelled by the Owner.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.17 Automated PO Review Screen 5.1.10 Low-Stock Alert Display
Field Validations	5.2.6 Data Description for Purchase Orders
Author	Zapanta, Zeus Levi; Albino, Erin Camille

Date	March 2026
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4.1.17 ANAIS-UC17 (View Supplier Dashboard)

ANAIS - UC17	View Supplier Dashboard
Description	This use case allows the Supplier to view the ANAIS Supplier Dashboard, which displays a summary of active purchase orders, pending confirmations, delivery statuses, and system notifications.
Actor	Supplier
Trigger	The supplier logs in and is redirected to the dashboard or navigates to it from the menu.
Complexity	Easy
Pre-condition	The Supplier must be logged in to the system.
Post condition	The Supplier can view the dashboard with a summary of active purchase orders, delivery statuses, and any pending notifications relevant to their account.

Basic flow	Actor Action	System Response
		Step 1. The system automatically displays the Supplier Dashboard upon successful login. Step 2. The dashboard shows a summary of active and pending purchase orders, recent delivery status updates, and any unread notifications based on the supplier's account
Alternate Flow	Actor Action	System Response
	Step 1. Supplier has no active purchase orders or notifications Step 2. System Delay or Loading Issue	Step 1a. The system displays an empty dashboard with a message indicating no current orders or activity for the supplier's account Step 2a. The system displays a loading indicator and retrieves data once the connection or processing is restored
Exception Flow	• If the system fails during dashboard loading, the Supplier is redirected to the login page with an error message.	

Includes	AN AIS-UC02 (Account Login)
Extends	• View Supplied Products (UC18) • View Orders from System (UC20) • View Delivery History (UC22)
Assumptions	• The Supplier's account is already created and managed by the Owner. • Inventory, order, and delivery data are already stored and updated in the database. • Data shown on the dashboard is assumed to be up to date at the time of login. • The system automatically loads the Supplier Dashboard after successful authentication.
Business Rules	• Dashboard content is role-based: the Supplier only sees data associated with their own account. • Pending purchase order confirmations are prominently displayed on the Supplier's dashboard. • Supplier accounts cannot be self-registered; only the Owner can create and deactivate supplier accounts.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.17 Supplier Dashboard Screen
Field Validations	5.2.6 Data Description for Purchase Orders
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.

Date	April 2026
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4.1.18 ANAIS-UC18 (View Supplied Products)

ANAIS - UC18	View Supplied Products
Description	This use case allows the Supplier to view the list of products they are registered to supply within the ANAIS system. The Supplier can review product names, descriptions, and any associated purchase order history for each product they supply.
Actor	Supplier
Trigger	The Supplier navigates to the Supplied Products section from the Supplier Dashboard.
Complexity	Easy
Pre-condition	• The Supplier must be logged in to the system. • The Supplier must be linked to at least one active product in the system. • The Supplier Dashboard (UC17) must be accessible.

Post condition	• A list of products associated with the Supplier's account is displayed. • The Supplier can view product details such as product name, description, category, and associated branch.	
Basic flow	Actor Action	System Response
	Step 1. The supplier navigates to the supplied products section from the dashboard. Step 3. The supplier review the displayed product list and may search or filter by product name or category	Step 2. The system queries the database and retrieves all products linked to the supplier's account Step 4. The system displays the filtered results based on the supplier's search or filter criteria Step 5. The system displays product details (name, description, category, branch) upon selection
Alternate Flow	Actor Action	System Response
	Step 2a. No products are linked to the supplier's account	Step 2a.1 The system displays an empty state with a message: "No products are currently associated with your account." The supplier is advised to contact the owner

	Step 3a. The supplier enters a search term that yields no results	Step 3a.1 The system displays a 'No results found' message and prompts the supplier to adjust the search term
Exception Flow	• If the system fails during product retrieval, an error message is displayed, and the Supplier is advised to retry or contact the Owner. • If the Supplier's session expires during browsing, they are redirected to the login page.	
Includes	ANAIS-UC17 (View Supplier Dashboard)	
Extends	• Update Supplied Product Information (UC19) • View Orders from System (UC20)	
Assumptions	• Products linked to a Supplier are managed by the Owner and cannot be added or removed by the Supplier. • The Supplier can only view products associated with their own account. • Product records are maintained and updated by the Owner or OIC.	

Business Rules	• Only products actively linked to the Supplier in the system are displayed. • Deactivated products remain visible in the Supplier's product list for historical reference but are visually flagged as inactive. • The Supplier cannot add, edit, or delete product records from this screen.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.18 Supplied Products Screen
Field Validations	5.2.4 Data Description for Products
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.
Date	April 2026

4.1.19 ANAIS-UC19 (Update Supplied Product Information)

ANAIS - UC19	Update Supplied Product Information
Description	This use case allows the Supplier to submit updates to supplied product information such as pricing, lead time, and availability status for products linked to their account. All updates submitted by the Supplier are flagged for review and must be approved by

	the Owner before changes are reflected in the system.	
Actor	Supplier	
Trigger	The Supplier identifies a need to update product details (e.g., change in price, lead time, or availability) and selects a product from the Supplied Products list.	
Complexity	Medium	
Pre-condition	• The Supplier must be logged in to the system. • The Supplier must be linked to the product they wish to update. • The Supplied Products list (UC18) must be displayed.	
Post condition	• The Supplier's proposed product update is submitted and flagged for Owner review. • The Owner receives a notification of the pending update request. • Existing product data remains unchanged until the Owner approves the update.	
Basic flow	Actor Action	System Response
	Step 1. The supplier selects a product from the Supplied Products list and clicks	Step 2. The system displays an update form pre- filled with current product information for the selected product.

	'Request Update'. Step 3. The Supplier modifies the relevant fields (e.g., unit price, lead time, availability status) and submits the form. System Response	Step 4. The system validates the submitted data for completeness and correct format. Step 5. The system saves the update request with a 'Pending Review' status and notifies the Owner via dashboard alert. Step 6. The system displays a confirmation message to the Supplier indicating the update request has been submitted for review.
Alternate Flow	Actor Action	System Response
	Step 4a. The Supplier submits incomplete or invalid data.	Step 4a.1 The system displays field-level validation errors. Step 4a.2 The system returns to the update form (Step 3) for correction.
	Step 3a. The Supplier cancels the update form without submitting. System Response	Step 3a.1 The system discards any changes and returns the Supplier to the Supplied Products list (UC18).

Exception Flow	• If the system fails during form submission, no update request is saved. The Supplier is advised to retry. • If the Supplier's session expires mid-form, they are redirected to the login page and unsaved data is discarded.
Includes	ANAIS-UC18 (View Supplied Products)
Extends	• Owner review and approval of supplier product update requests
Assumptions	• Suppliers can only request updates to their own linked products. • The Owner retains final authority over all product data in the system. • Update requests are logged for audit purposes regardless of approval outcome. • The Supplier is notified once the Owner approves or rejects their update request.
Business Rules	• Supplier-submitted updates do not take effect until approved by the Owner. • Only one pending update request per product per Supplier may exist at a time. • The Owner may approve, reject, or request further revision of a submitted update. • Editable fields are limited to: unit price, lead time (days), and availability status. • Core product attributes (product name, category, branch assignment) cannot be modified by the Supplier.

Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.19 Supplier Product Update Request Form
Field Validations	5.2.4 Data Description for Products; 5.2.5 Data Description for Supplier Records
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.
Date	April 2026

4.1.20 ANAIS-UC20 (View Orders from System)

ANAIS - UC20	View Orders from System
Description	This use case allows the Supplier to view all purchase orders directed to them from the ANAIS system, including both automated and manually generated purchase orders. The Supplier can review order details such as ordered products, quantities, branch, and order status. This view also supports confirming pending purchase orders.
Actor	Supplier
Trigger	The Supplier navigates to the Orders section from the Supplier Dashboard, or receives a notification of a new purchase order.

Complexity	Easy	
Pre-condition	• The Supplier must be logged in to the system. • At least one purchase order must exist for the Supplier in the system. • The Supplier Dashboard (UC17) must be accessible.	
Post condition	• The Supplier views a list of all purchase orders associated with their account, filtered by status if applicable. • The Supplier may proceed to confirm a pending purchase order (UC21) from this view.	
Basic flow	Actor Action	System Response
	Step 1. The Supplier navigates to the Orders section from the Supplier Dashboard. Step 3. The Supplier reviews the order list and may apply status filters (e.g., Pending, Confirmed, Partially Delivered, Fully	Step 2. The system retrieves all purchase orders linked to the Supplier from the database, sorted by date (most recent first). Step 4. The system displays the filtered purchase order list. Step 5. The system displays full order details (product name, quantity, branch, order date, status, and PO number) upon the Supplier

	Delivered, Cancelled). Step 6. The Supplier reviews the order details and may choose to confirm the order (UC21) or proceed to update the delivery status (UC21).	selecting a specific order.
Alternate Flow	Actor Action	System Response
	Step 2a. No purchase orders exist for the Supplier	Step 2a.1 The system displays an empty state with the message: 'No orders have been placed for your account yet.'
	Step 3a. The Supplier applies a filter that returns no results.	Step 3a.1 The system displays a 'No results found for the selected filter' message and prompts the Supplier to adjust filter criteria.
Exception Flow	• If the system fails during order retrieval, an error message is displayed and the Supplier is advised to retry. • If the Supplier's session expires, they are redirected to the login page.	

Includes	AN AIS-UC17 (View Supplier Dashboard)
Extends	• Confirm Purchase Order: Supplier acknowledges and accepts a pending purchase order. • Update Order Status (UC21): Supplier updates the delivery status of a confirmed order. • Receive Automated Purchase Order Notifications: The Supplier receives automated email notifications when a new PO is generated.
Assumptions	• Both automated (UC16) and manually recorded (UC14) purchase orders are visible to the Supplier. • The Supplier can only view orders associated with their own account. • Order data is read-only from this screen except for confirmation and status update actions.
Business Rules	• Suppliers may only view, confirm, or update the status of purchase orders linked to their own account. • Purchase orders with a status of 'Draft' are not visible to the Supplier until confirmed by the Owner and set to 'Pending'. • Cancelled purchase orders remain visible to the Supplier for reference but cannot be reactivated. • Order data is not editable by the Supplier; only confirmation and delivery status actions are permitted.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.20 Supplier Order View Screen • 5.1.14 Purchase Order Management Form

Field Validations	5.2.6 Data Description for Purchase Orders
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.
Date	April 2026

4.1.21 ANAIS-UC21 (Update Order Status)

ANAIS - UC21	Update Order Status
Description	This use case allows the Supplier to acknowledge and confirm a pending purchase order and/or update the delivery status of an active order. This includes confirming the purchase order upon receipt, marking items as dispatched, and updating partial or full delivery completion. These updates are reflected in the system and visible to the Owner and OIC.
Actor	Supplier

Trigger	The Supplier receives a purchase order notification or navigates to a specific order in the Orders section (UC20) and initiates a status update.	
Complexity	Medium	
Pre-condition	• The Supplier must be logged in to the system. • A purchase order with a status of 'Pending', 'Confirmed', or 'Partially Delivered' must exist for the Supplier. • The Supplier has accessed the order details via UC20 (View Orders from System).	
Post condition	• The purchase order status is updated to reflect the Supplier's action (e.g., Confirmed, Dispatched, Partially Delivered, Fully Delivered). • The Owner and OIC are notified of the status change via dashboard notification. • The updated status is logged with a timestamp and the Supplier's identity.	
Basic flow	Actor Action	System Response
	Step 1. The Supplier selects a purchase order with 'Pending' status from the Orders list.	Step 2. The system displays the full purchase order details including products, quantities, and current status.

	Step 3. The Supplier clicks 'Confirm Order' to acknowledge and accept the purchase order. Step 6. At a later stage, the Supplier selects the confirmed order and updates the delivery status (e.g., 'Dispatched', 'Partially Delivered', 'Fully Delivered').	Step 4. The system updates the purchase order status to 'Confirmed' and records the confirmation timestamp. Step 5. The system sends a dashboard notification to the Owner and OIC that the order has been confirmed by the Supplier. Step 7. The system validates the status update and saves the new delivery status with a timestamp. Step 8. The system notifies the Owner and OIC of the updated delivery status.
Alternate Flow	Actor Action	System Response
	Step 3a. The Supplier declines the purchase order.	Step 3a.1 The system updates the order status to 'Cancelled' and notifies the Owner and OIC. Step 3a.2 The system prompts the Owner to manually create a new purchase order or assign an alternative supplier.

	Step 6a. The Supplier selects 'Partially Delivered' and specifies which items have been delivered and their quantities.	Step 6a.1 The system updates the order status to 'Partially Delivered' and records the partial delivery details. Step 6a.2 The system notifies the Owner and OIC and keeps the order open for subsequent delivery updates.
Exception Flow	• If the system fails during status update, the order status remains unchanged and an error message is displayed. The Supplier is advised to retry. • If the Supplier's session expires before completing the status update, they are redirected to the login page and no changes are saved. • If the purchase order has already been cancelled by the Owner, the Supplier is shown an error message and the status update action is not allowed.	
Includes	AN AIS-UC20 (View Orders from System)	
Extends	• Record Supplier Delivery (UC15): Once delivery status is marked as 'Fully Delivered', the Owner or OIC may record the formal supplier delivery in the system. • Receive Automated Purchase Order Notifications: Supplier receives notification upon a new PO being assigned to them.	

Assumptions	• The Supplier acts in good faith when confirming or updating order statuses. • Delivery status updates from the Supplier are informational and do not automatically update inventory levels; this is done by the Owner or OIC via UC15. • The Supplier may update the delivery status multiple times for partially delivered orders until the order is fully delivered or cancelled.
Business Rules	• Only orders with a status of 'Pending' can be confirmed or declined by the Supplier. • Only orders with a status of 'Confirmed' or 'Partially Delivered' can have their delivery status updated. • The Supplier may not revert a 'Fully Delivered' status once set. • All status updates are logged with a timestamp and the Supplier's account ID for audit purposes. • Status changes follow the lifecycle: Pending → Confirmed → Dispatched → Partially Delivered → Fully Delivered.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.21 Order Status Update Screen • 5.1.15 Supplier Delivery Recording Screen
Field Validations	5.2.6 Data Description for Purchase Orders; 5.2.7 Data Description for Delivery Records
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.
Date	April 2026

4.1.22 ANAIS-UC22 (View Delivery History)

ANAIS - UC22	View Delivery History
Description	This use case allows the Supplier to access a complete history of all deliveries made against purchase orders associated with their account. The Supplier can view delivery records including products delivered, quantities, dates, and current delivery status for both past and ongoing orders.
Actor	Supplier
Trigger	The Supplier navigates to the Delivery History section from the Supplier Dashboard or from the Order Details view.
Complexity	Easy
Pre-condition	• The Supplier must be logged in to the system. • At least one purchase order with a delivery record must exist for the Supplier. • The Supplier Dashboard (UC17) must be accessible.

Post condition	• The Supplier views a list of all delivery records associated with their purchase orders. • The Supplier can review delivery details including products, quantities, delivery dates, and status.	
Basic flow	Actor Action	System Response
	Step 1. The Supplier navigates to the Delivery History section from the dashboard or from a specific order detail view. Step 3. The Supplier reviews the delivery history list and may filter by date range or delivery status.	Step 2. The system retrieves all delivery records associated with the Supplier's purchase orders from the database. Step 4. The system displays the filtered delivery history based on the Supplier's selected criteria. Step 5. The system displays full delivery details (PO number, products delivered, quantities, delivery date, receiving branch, and status) when the Supplier selects a record.
Alternate Flow	Actor Action	System Response
	Step 2a. No delivery records exist	Step 2a.1 The system displays an

	for the Supplier's account.	empty state with the message: 'No delivery records found for your account.'
	Step 3a. The Supplier applies a date or status filter that returns no results.	Step 3a.1 The system displays a 'No results found for the selected filters' message and prompts the Supplier to adjust the filter criteria.
Exception Flow	• If the system fails during delivery record retrieval, an error message is displayed and the Supplier is advised to retry. • If the Supplier's session expires during browsing, they are redirected to the login page.	
Includes	ANAIIS-UC17 (View Supplier Dashboard)	
Extends	Update Order Status (UC21): The Supplier may navigate directly to update the delivery status of an active order from the delivery history view.	
Assumptions	• Delivery records displayed are those formally recorded by the Owner or OIC via UC15 (Record Supplier Delivery) as well as status updates logged by the Supplier via UC21. • The Supplier can only access delivery records associated with their own account. • Delivery history data is read-only for the Supplier.	

Business Rules	• Suppliers may only view delivery records linked to their own purchase orders. • Delivery records cannot be edited or deleted by the Supplier. • Records are sorted in descending chronological order by default. • Delivery records remain permanently visible in the Supplier's history, including those associated with cancelled orders. • The delivery history must reflect both Supplier-logged status updates (UC21) and Owner/OIC-recorded deliveries (UC15) for a complete audit trail.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.22 Supplier Delivery History Screen • 5.1.15 Supplier Delivery Recording Screen
Field Validations	5.2.7 Data Description for Delivery Records
Author	Zapanta, Zeus Levi; Lopez, Hans Jayden A.
Date	April 2026

4.1.23 ANAIS-UC23 (Delete User Account)

ANAIS - UC23	Delete User Account
Description	This use case allows the Owner to permanently remove a user account from the system. All associated access credentials are revoked and the account is no longer

	retrievable, while audit logs are retained for record-keeping purposes.	
Actor	Owner	
Trigger	The Owner needs to permanently remove a user account that is no longer required in the system (e.g., a staff member who has left the business).	
Complexity	Easy	
Pre-condition	• The Owner must be logged in to the system. • The Account Management page must be open and displaying the user list. • The target user account must exist in the system.	
Post condition	• The selected user account is permanently deleted. All associated access credentials are revoked and the account record is no longer retrievable. Audit logs are retained for record-keeping.	
Basic flow	Actor Action	System Response
	Step 1. Owner navigates to the Account Management page and selects the user account to be deleted. Step 3.	Step 2. System displays the selected user's account details and presents the Delete Account option. Step 4. System displays a confirmation dialog

	Owner selects the delete option. Step 5. Owner confirms the deletion	warning the Owner that deletion is permanent and cannot be undone. Step 6. System permanently removes the user account and all associated credentials from the database, retains audit logs, and notifies the Owner that the deletion was successful.
Alternate Flow	Actor Action	System Response
	Step 5a. Owner chooses to cancel the deletion at the confirmation dialog.	Step 5a.1 System dismisses the confirmation dialog and returns to the Account Management page without modifying the account.
Exception Flow	• If an error occurs during the deletion process (e.g., database failure, server error, or network interruption): • The system shall display an error message indicating that the deletion was unsuccessful. • The user account shall remain unchanged in the system. • The Owner may be logged out and redirected to the login page if a critical session failure occurs.	
Includes	ANAIS-UC01 (Account Creation) ANAIS-UC02 (Account Login) ANAIS-UC03 (Logout)	

Extends	
Assumptions	- Deleted accounts cannot be restored; this operation is irreversible. - Audit logs associated with the deleted account are retained in the system for historical reference. - The Owner's own account cannot be deleted through this use case.
Business Rules	- Only the Owner has the privilege to permanently delete user accounts. - Deletion is permanent and cannot be undone. - All associated access credentials are revoked immediately upon deletion. - A confirmation prompt must be acknowledged before deletion proceeds. - Audit logs must be retained even after account deletion.
Related Models/ Diagrams	4.2 System Use Case Diagram 5.1.22 Supplier Delivery History Screen 5.1.15 Supplier Delivery Recording Screen
Field Validations	4.2 System Use Case Diagram 5.1.1 Account Management Screen 5.2.1 Data Description for Account Management
Author	Lopez, Hans Jayden A.
Date	May 2026

4.1.25 ANAIS-UC25 (Deactivate User Account)

ANAIS - UC25	Deactivate User Account	
Description	This use case allows the Owner to deactivate a user account from the system, revoking the user's access while retaining their account record and audit logs for record-keeping purposes.	
Actor	Owner	
Trigger	The Owner needs to revoke system access for a specific user (OIC or Employee) without permanently deleting their account record.	
Complexity	Easy	
Pre-condition	• The Owner must be logged in to the system. • The Account Management page must be open and displaying the user list. • The target user account must exist and currently be in an active state.	
Post condition	• The selected user account is deactivated. The account record and associated audit logs are retained in the system. The deactivated user is no longer able to log in.	
Basic flow	Actor Action	System Response

	Step 1. Owner navigates to the Account Management page and selects the user account to deactivate. Step 3. Owner selects the deactivate option. Step 5. Owner confirms the deactivation.	Step 2. System displays the selected user's account details and presents the Deactivate Account option. Step 4. System displays a confirmation dialog prompting the Owner to confirm the deactivation. Step 6. System sets the user account status to Inactive, revokes login access, retains the account record and audit logs, and notifies the Owner of the successful deactivation.
Alternate Flow	Actor Action	System Response
	Step 5a. Owner chooses to cancel the deactivation at the confirmation dialog.	Step 5a.1 System dismisses the confirmation dialog and returns to the Account Management page without modifying the account.
Exception Flow	• If an error occurs during the deactivation process (e.g., database connection failure, server error): • The system shall display an error message indicating that the deactivation was unsuccessful. • The user account shall remain in its current active state.	

	• The Owner may be logged out and redirected to the login page if a critical session failure occurs.
Includes	ANAIIS-UC01 (Account Creation) ANAIIS-UC02 (Account Login) ANAIIS-UC03 (Logout)
Extends	
Assumptions	• A deactivated account can be reactivated by the Owner at a later time. • Deactivated accounts are retained in the system for audit and historical record-keeping. • The Owner's own account cannot be deactivated through this use case.
Business Rules	• Only the Owner has the privilege to deactivate user accounts. • Deactivated users cannot log in to the system. • Account records and audit logs must be preserved upon deactivation. • A confirmation prompt must be acknowledged before deactivation proceeds.
Related Models/ Diagrams	• 4.2 System Use Case Diagram • 5.1.22 Supplier Delivery History Screen • 5.1.15 Supplier Delivery Recording Screen

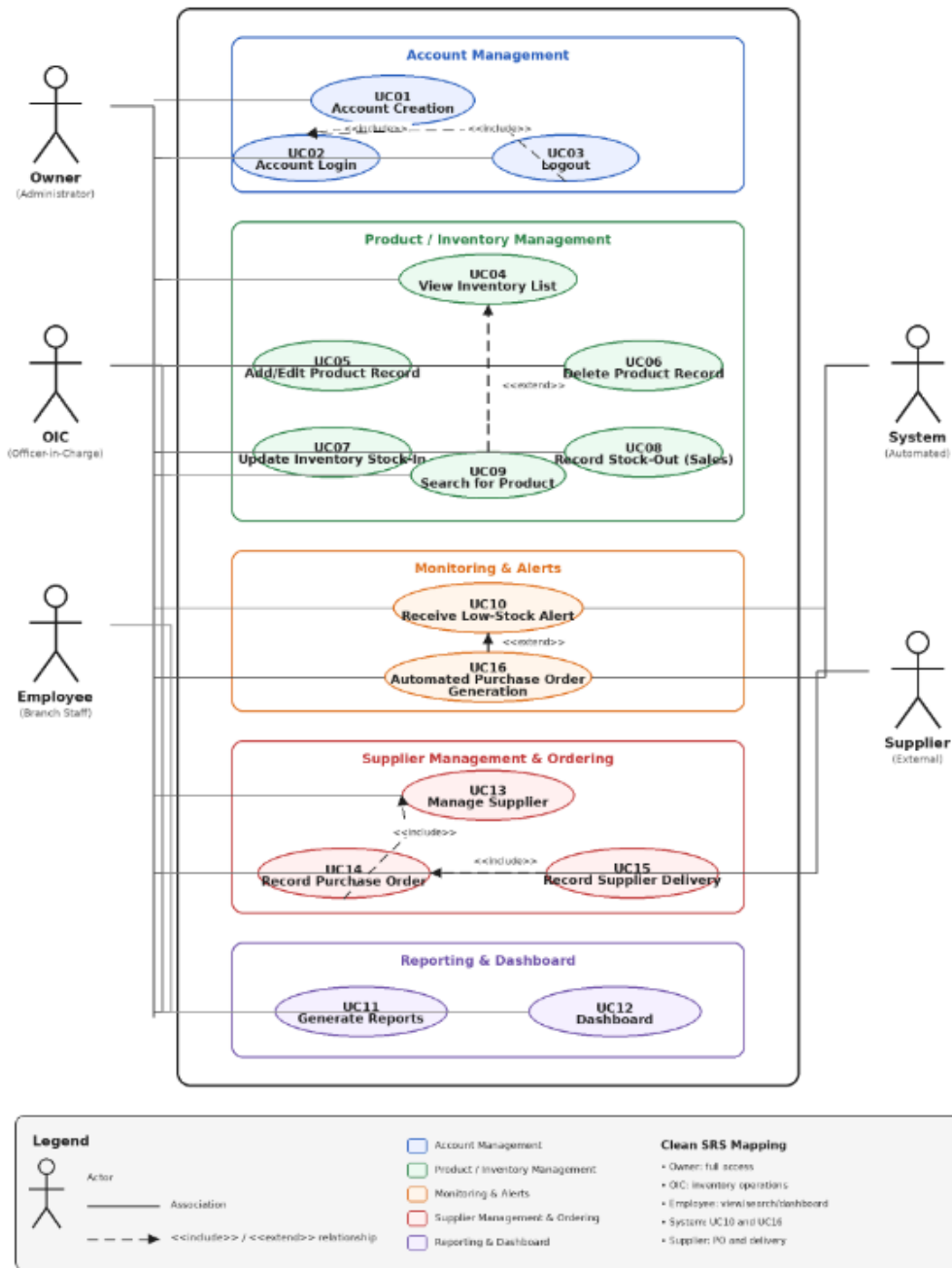
Field Validations	4.2 System Use Case Diagram 5.1.1 Account Management Screen 5.2.1 Data Description for Account Management
Author	Lopez, Hans Jayden A.
Date	May 2026

4.2 Use Case Diagram

The following Use Case Diagram illustrates the interactions between the different user roles (Owner, OIC, Employee) and the ANAIS system. It provides a high-level view of all system functionalities and the actors involved.

ANAIIS — Autobox and Autophoria Inventory System

4.2 System Use Case Diagram



5 Other Requirements

5.1 Interfaces

This section describes the screen forms and user interface components for each major function of ANAIS.

5.1.1 Account Creation Form

This form is used by the Owner to create new user accounts for OICs and Employees.

Field Name	Type	Required	Description / Validation
Full Name	Text	Required	Employee's full name; max 100 characters
Role	Dropdown	Required	Select: Owner, OIC, or Employee
Branch	Dropdown	Required	Select: Autobox or Autophoria
Username	Text	Required	Unique username; max 50 characters; alphanumeric
Password	Password	Required	Minimum 8 characters; must include

			letters and numbers
Confirm Password	Password	Required	Must match Password field

5.1.2 Login Form

This form allows all users to authenticate and access the system based on their role.

Field Name	Type	Required	Description / Validation
Username	Text	Required	Registered username
Password	Password	Required	Account password

5.1.3 Logout

The logout function is accessible from the navigation menu. A single click on the Logout button ends the session and redirects the user to the login page. No form input is required.

5.1.4 Inventory View

This screen displays the list of all products per branch with their current stock levels and status.

Field Name	Type	Required	Description / Validation
Branch Filter	Dropdown	Optional	Filter by Autobox, Autophoria, or All Branches
Category Filter	Dropdown	Optional	Filter by product category
Search Bar	Text	Optional	Search by product name or SKU
Product Name	Text	Display	Name of the product
SKU	Text	Display	Stock Keeping Unit identifier
Category	Text	Display	Product category
Quantity	Number	Display	Current stock quantity

Status	Text	Display	In Stock / Low Stock / Out of Stock
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5.1.5 Product Management Form

This form is used by the Owner to add new products or edit existing product records.

Field Name	Type	Required	Description / Validation
Product Name	Text	Required	Name of the product; max 150 characters
SKU	Text	Required	Unique stock keeping unit code
Category	Dropdown	Required	Product category (e.g., Audio, Lighting, Interior, etc.)
Branch	Dropdown	Required	Autobox or Autophoria
Price	Decimal	Required	Selling price in Philippine Peso (PHP)

Quantity	Number	Required	Initial stock quantity; must be ≥ 0
Low-Stock Threshold	Number	Required	Minimum quantity before alert is triggered
Description	Text Area	Optional	Additional product notes or description

5.1.6 Delete Confirmation

This is a confirmation dialog displayed before a product record is permanently deleted. It shows the product name and asks the Owner to confirm or cancel.

5.1.7 Stock-In Form

This form is used by the OIC to record incoming products upon delivery.

Field Name	Type	Required	Description / Validation
Product	Dropdown/Search	Required	Select product from existing inventory
Quantity Received	Number	Required	Number of units received; must be > 0

Delivery Date	Date	Required	Date of delivery
Remarks	Text Area	Optional	Additional notes about the delivery

5.1.8 Stock-Out Form

This form is used by the Owner or OIC to record product sales and deduct from inventory.

Field Name	Type	Required	Description / Validation
Product	Dropdown/Search	Required	Select product from existing inventory
Quantity Sold	Number	Required	Number of units sold; must be > 0 and <= available stock
Sale Date	Date	Required	Date of the sale transaction
Remarks	Text Area	Optional	Additional notes about the transaction

5.1.9 Search Product

This search panel allows users to quickly locate products across the inventory.

Field Name	Type	Required	Description / Validation
Search Term	Text	Optional	Enter product name or SKU
Category	Dropdown	Optional	Filter by product category
Branch	Dropdown	Optional	Filter by branch (Owner only sees all branches)

5.1.10 Low-Stock Alert Display

This is a notification panel displayed on the dashboard when one or more products fall below their defined low-stock threshold.

Field Name	Type	Required	Description / Validation
Product Name	Text	Display	Name of the low-stock product

Current Quantity	Number	Display	Remaining stock quantity
Low-Stock Threshold	Number	Display	Defined minimum quantity
Branch	Text	Display	Branch where the alert was triggered

5.1.11 Generate Reports

This screen allows the Owner and OIC to configure and generate inventory reports.

Field Name	Type	Required	Description / Validation
Branch	Dropdown	Required	Select: Autobox, Autophoria, or All Branches
Date Range - From	Date	Required	Start date for the report period
Date Range - To	Date	Required	End date for the report period

Category	Dropdown	Optional	Filter by product category
Report Type	Dropdown	Required	Select: Stock Summary, Sales Report, or Low-Stock Report

5.1.12 Dashboard

The dashboard is the main screen displayed after login. It provides a real-time overview of inventory status for the logged-in user's branch or both branches (Owner only).

Field Name	Type	Required	Description / Validation
Stock Summary	Display Panel	Auto	Total products, total stock value per branch
Low-Stock Alerts	Notification Panel	Auto	List of products below threshold
Branch Overview	Display Panel	Auto	Inventory snapshot per branch
Quick Links	Navigation Buttons	Auto	Quick access to key functions

5.1.13 Landing Page

The landing page is the first screen displayed when the user accesses ANAIS. It presents the system name, logo, and a prompt to log in.

5.1.14 Supplier Management Form

This form is used by the Owner to add new suppliers or edit existing supplier records. Suppliers must be registered before they can be selected in purchase orders.

Field Name	Type	Required	Description / Validation
Supplier Name	Text	Required	Full business name of the supplier; max 150 characters; must be unique
Contact Person	Text	Required	Name of the primary contact person; max 100 characters
Contact Number	Text	Required	Phone or mobile number; numeric format; max 15 digits
Email Address	Email	Optional	Valid email address; must follow standard email format

Address	Text Area	Required	Complete business address of the supplier
Status	Toggle/Radio	Required	Active or Inactive; defaults to Active on creation
Remarks	Text Area	Optional	Additional notes about the supplier

5.1.15 Purchase Order Form

This form is used by the Owner to create and record a purchase order (PO) for products from a registered supplier.

Field Name	Type	Required	Description / Validation
PO Number	Text	Auto-generated	Unique PO number assigned by the system upon saving
Supplier	Dropdown	Required	Select from active registered suppliers
Branch	Dropdown	Required	Select: Autobox or Autophoria -

			the receiving branch
Order Date	Date	Auto	Automatically set to the current date upon PO creation
Expected Delivery Date	Date	Required	Expected date of delivery from the supplier
Product (Line Item)	Dropdown/Search	Required	Select product from inventory; multiple line items allowed
Quantity Ordered	Number	Required	Number of units to order per line item; must be > 0
Unit Cost	Decimal	Required	Cost per unit from supplier in Philippine Peso (PHP)
Total Cost	Decimal	Auto-computed	Auto-computed: Quantity Ordered x Unit Cost per line item

PO Status	Text	Auto	Automatically set to Pending upon creation
Remarks	Text Area	Optional	Additional notes or instructions for the supplier

5.1.16 Supplier Delivery Form

This form is used by the Owner or OIC to record the actual receipt of products from a supplier, referencing an existing purchase order.

Field Name	Type	Required	Description / Validation
PO Reference	Dropdown/Search	Required	Select from pending or partially delivered purchase orders
Delivery Date	Date	Required	Actual date the products were received at the branch
Product (Line Item)	Display	Auto	Populated from the selected PO; shows ordered products and quantities

Quantity Ordered	Number	Display	Ordered quantity per line item from the referenced PO
Quantity Received	Number	Required	Actual units received; must be > 0 and <= quantity ordered
Delivery Receipt No.	Text	Optional	Reference number from the supplier delivery receipt
Received By	Text	Auto	Automatically recorded as the logged-in user name
Remarks	Text Area	Optional	Additional notes about the delivery

5.1.17 Automated PO Review Screen

This screen is displayed to the Owner when the system has automatically generated a draft purchase order following a low-stock alert. It allows the Owner to review the system-generated draft, adjust details if necessary, and either confirm or cancel the draft PO. This screen is accessible from the low-stock alert notification on the dashboard.

Field Name	Type	Required	Description / Validation
Draft PO Number	Text	Auto-generated	System-assigned draft identifier; replaced by a permanent PO number upon confirmation
Triggered By	Text	Auto	Name of the product that triggered the low-stock alert and initiated the draft PO
Current Stock Qty	Number	Display	Current stock quantity of the triggering product at the time of alert
Low-Stock Threshold	Number	Display	Defined minimum quantity that triggered the automated generation
Supplier	Dropdown	Required	Pre-filled with the product linked supplier; Owner may change to another active

			supplier if needed
Branch	Text	Auto	Pre-filled with the branch where the low-stock alert was triggered
Product (Line Item)	Display	Auto	Pre-filled with the triggering product; additional products may be added by the Owner
Suggested Qty	Number	Auto	System-suggested reorder quantity (default: 2x the low-stock threshold); editable by the Owner
Unit Cost	Decimal	Required	Cost per unit from the supplier in Philippine Peso (PHP); must be entered or confirmed by the Owner
Expected Delivery Date	Date	Required	Expected date of delivery; must be set by

			the Owner before confirmation
PO Status	Text	Auto	Set to Draft until Owner confirms; changes to Pending upon confirmation
Remarks	Text Area	Optional	Additional notes or instructions for the purchase order
Confirm Button	Action	Required	Owner confirms the draft PO; system saves it as a Pending purchase order with a unique PO number
Cancel Button	Action	Optional	Owner cancels and discards the draft PO; low-stock alert remains on the dashboard

5.2 Logical Data Design

This section describes the data elements associated with each major use case and system function in ANAIS.

5.2.1 Data Description for Account Management

Describes the data elements used for creating and managing user accounts in ANAIS.

Data Element	Data Type	Constraints	Description
user_id	INT	Primary Key, Auto-increment	Unique identifier for each user account
full_name	VARCHAR(100)	NOT NULL	Full name of the user
username	VARCHAR(50)	NOT NULL, Unique	Login username; must be unique across the system
password_hash	VARCHAR(255)	NOT NULL	Hashed password stored securely

role	ENUM	NOT NULL	Values: Owner, OIC, Employee
branch	VARCHAR(50)	NOT NULL	Assigned branch: Autobox or Autophoria
is_active	BOOLEAN	Default: TRUE	Indicates if the account is active
created_at	DATETIME	NOT NULL	Timestamp when the account was created
updated_at	DATETIME	NOT NULL	Timestamp of the last account update

5.2.2 Data Description for Inventory Records

Describes the data elements for product records stored in the ANAIS inventory database.

Data Element	Data Type	Constraints	Description
product_id	INT	Primary Key, Auto-increment	Unique identifier for each product
product_name	VARCHAR(150)	NOT NULL	Name of the product

sku	VARCHAR(50)	NOT NULL, Unique	Stock Keeping Unit; unique product code
category	VARCHAR(100)	NOT NULL	Product category (e.g., Audio, Lighting, Interior)
branch	VARCHAR(50)	NOT NULL	Branch where the product is assigned
price	DECIMAL(10,2)	NOT NULL	Selling price in Philippine Peso
quantity	INT	NOT NULL, >= 0	Current stock quantity
low_stock_thres hold	INT	NOT NULL, > 0	Minimum quantity before a low-stock alert is triggered
description	TEXT	Nullable	Optional product notes or description

created_at	DATETIME	NOT NULL	Timestamp when the product was added
updated_at	DATETIME	NOT NULL	Timestamp of the last product update

5.2.3 Data Description for Stock Transactions

Describes the data elements for stock-in and stock-out transactions recorded in ANAIS.

Data Element	Data Type	Constraints	Description
transaction_id	INT	Primary Key, Auto-increment	Unique identifier for each transaction
product_id	INT	Foreign Key -> product_id	Reference to the product involved in the transaction
transaction_type	ENUM	NOT NULL	Values: Stock-In, Stock-Out
quantity	INT	NOT NULL, > 0	Number of units involved

			in the transaction
branch	VARCHAR(50)	NOT NULL	Branch where the transaction occurred
transaction_date	DATE	NOT NULL	Date of the transaction
performed_by	INT	Foreign Key -> user_id	Reference to the user who recorded the transaction
remarks	TEXT	Nullable	Optional notes about the transaction
created_at	DATETIME	NOT NULL	Timestamp when the transaction was recorded

5.2.4 Data Description for Report Generation

Describes the data elements and filter parameters used when generating inventory reports in ANAIS.

Data Element	Data Type	Constraints	Description
report_id	INT	Primary Key, Auto-increment	Unique identifier for

			each generated report
report_type	ENUM	NOT NULL	Values: Stock Summary, Sales Report, Low-Stock Report
branch	VARCHAR(50)	NOT NULL	Branch scope: Autobox, Autophoria, or All Branches
date_from	DATE	NOT NULL	Start date of the report period
date_to	DATE	NOT NULL	End date of the report period
category_filter	VARCHAR(100)	Nullable	Optional category filter applied to the report
generated_by	INT	Foreign Key -> user_id	Reference to the user who generated the report

generated_at	DATETIME	NOT NULL	Timestamp when the report was generated
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5.2.5 Data Description for Supplier Records

Describes the data elements for supplier records stored in the ANAIS database.

Data Element	Data Type	Constraints	Description
supplier_id	INT	Primary Key, Auto-increment	Unique identifier for each supplier
supplier_name	VARCHAR(150)	NOT NULL, Unique	Full business name of the supplier
contact_person	VARCHAR(100)	NOT NULL	Name of the supplier primary contact person
contact_number	VARCHAR(15)	NOT NULL	Phone or mobile number of the supplier
email	VARCHAR(150)	Nullable	Email address of the supplier

address	TEXT	NOT NULL	Complete business address of the supplier
status	ENUM	NOT NULL, Default: Active	Values: Active, Inactive
remarks	TEXT	Nullable	Optional notes about the supplier
created_by	INT	Foreign Key -> user_id	Reference to the user who created the record
created_at	DATETIME	NOT NULL	Timestamp when the supplier record was created
updated_at	DATETIME	NOT NULL	Timestamp of the last update to the supplier record

5.2.6 Data Description for Purchase Orders

Describes the data elements for purchase orders created in ANAIS when ordering products from suppliers.

Data Element	Data Type	Constraints	Description
po_id	INT	Primary Key, Auto-increment	Unique identifier for each purchase order
po_number	VARCHAR(50)	NOT NULL, Unique	System-generated unique purchase order number
supplier_id	INT	Foreign Key -> supplier_id	Reference to the supplier fulfilling the order
branch	VARCHAR(50)	NOT NULL	Branch receiving the ordered products
order_date	DATE	NOT NULL	Date the purchase order was created
expected_delivery_date	DATE	NOT NULL	Expected date of delivery from the supplier

po_status	ENUM	NOT NULL, Default: Pending	Values: Pending, Partially Delivered, Fully Delivered, Cancelled
remarks	TEXT	Nullable	Optional notes or instructions for the purchase order
created_by	INT	Foreign Key -> user_id	Reference to the Owner who created the PO
created_at	DATETIME	NOT NULL	Timestamp when the PO was created
updated_at	DATETIME	NOT NULL	Timestamp of the last update to the PO

5.2.7 Data Description for Supplier Deliveries

Describes the data elements for supplier delivery records in ANAIS, linked to purchase orders and inventory stock updates.

Data Element	Data Type	Constraints	Description
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delivery_id	INT	Primary Key, Auto-increment	Unique identifier for each delivery record
po_id	INT	Foreign Key -> po_id	Reference to the associated purchase order
product_id	INT	Foreign Key -> product_id	Reference to the delivered product
quantity_order ed	INT	NOT NULL, > 0	Quantity of the product ordered in the PO line item
quantity_receiv ed	INT	NOT NULL, > 0	Actual quantity of the product received in this delivery
delivery_date	DATE	NOT NULL	Actual date the products were received at the branch
delivery_receip t_no	VARCHAR(100)	Nullable	Reference number from the supplier delivery receipt

received_by	INT	Foreign Key -> user_id	Reference to the user who recorded the delivery
remarks	TEXT	Nullable	Optional notes about the delivery
created_at	DATE TIME	NOT NULL	Timestamp when the delivery record was created